

Sheet #		Content		Sheet #		Content	

Revision History

REV	DESCRIPTION
A	EVT2.0 Initial release

Board Change History

SCHEMATIC MCN	HW VERSION	REVISION	DESCRIPTION OF CHANGE
LD20-P3555	EVT1.1	A	EVT1.1 pre-liminary release
LD20-P4374	EVT2.0	A	EVT2.0 Initial release, the details refer to EVT2.0 change list in "Change list" page

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Revision History

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Title

Sheet

Size

Date

Rev

RF Topology

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Block Diagram

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Title		
Drawn	By	Rev
Date		
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MSM8953 GPIO Configuration For QRD8953

GPIO_0	NFC_SPI_ESE_MOSI	GPIO_41		GPIO_82	FM_DATA
GPIO_1	NFC_SPI_ESE_MISO	GPIO_42	ACCL_INT1	GPIO_83	BT_CTL
GPIO_2	NFC_SPI_ESE_CS_N	GPIO_43	ALSP_INT_N	GPIO_84	BT_DATA
GPIO_3	NFC_SPI_ESE_CLK	GPIO_44	MAG_DRDY_INT	GPIO_85	KEY_VOL_UP_N
GPIO_4	UART_MSM_TX	GPIO_45	GYRO_INT	GPIO_86	
GPIO_5	UART_MSM_RX	GPIO_46	SCAM_D_ELD0_EN	GPIO_87	
GPIO_6		GPIO_47	SENSOR_SPI_CS2_N	GPIO_88	
GPIO_7		GPIO_48	FP_INT_N	GPIO_89	
GPIO_8		GPIO_49	UIM_BATT_ALARM	GPIO_90	
GPIO_9		GPIO_50	WCD_ELD0_EN	GPIO_91	
GPIO_10	TP_I2C_SDA	GPIO_51	UIM1_DATA	GPIO_92	
GPIO_11	TP_I2C_SCL	GPIO_52	UIM1_CLK	GPIO_93	SCAM_ID
GPIO_12		GPIO_53	UIM1_RESET	GPIO_94	
GPIO_13		GPIO_54		GPIO_95	
GPIO_14	SENSOR_I2C_SDA	GPIO_55	UIM2_DATA	GPIO_96	WSA_EN
GPIO_15	SENSOR_I2C_SCL	GPIO_56	UIM2_CLK	GPIO_97	
GPIO_16	NFC_DISABLE	GPIO_57	UIM2_RESET	GPIO_98	
GPIO_17	NFC_IRQ	GPIO_58		GPIO_99	
GPIO_18	NFC_I2C_SDA	GPIO_59		GPIO_100	GRFC0_SEL
GPIO_19	NFC_I2C_SCL	GPIO_60	GP_PDM_A0	GPIO_101	GRFC1_SEL
GPIO_20	SENSOR_SPI_MOSI	GPIO_61	LCD_RST_N	GPIO_102	GRFC2_SEL
GPIO_21	SENSOR_SPI_MISO	GPIO_62	NFC_DWL_REQ	GPIO_103	GRFC3_SEL
GPIO_22	SENSOR_SPI_CS0_N	GPIO_63		GPIO_104	GRFC4_SEL
GPIO_23	SENSOR_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	GRFC5_SEL
GPIO_24	LCD_TE0	GPIO_65	TP_INT_N	GPIO_106	WDOG_DISABLE
GPIO_25		GPIO_66		GPIO_107	GRFC7_SEL
GPIO_26	CAM_MCLK0	GPIO_67	WCD_RST_N	GPIO_108	GRFC8_SEL
GPIO_27	CAM_MCLK1	GPIO_68		GPIO_109	GRFC9_SEL
GPIO_28		GPIO_69	WCD_MSM_MCLK	GPIO_110	
GPIO_29	CAM_I2C_SDA0	GPIO_70	SLIMBUS_CLK	GPIO_111	
GPIO_30	CAM_I2C_SCL0	GPIO_71	SLIMBUS_DATA0	GPIO_112	
GPIO_31	CAM_I2C_SDA1	GPIO_72		GPIO_113	
GPIO_32	CAM_I2C_SDA1	GPIO_73	WCD_INTR1	GPIO_114	
GPIO_33		GPIO_74	WCD_INTR2	GPIO_115	
GPIO_34	FLASH_STROBE_NOW	GPIO_75	BT_SSB1	GPIO_116	EXT_GPS_LNA_EN
GPIO_35		GPIO_76	WL_CMD_DATA_2	GPIO_117	CH0_GSM_TX_PHASE_D0
GPIO_36		GPIO_77	WL_CMD_DATA_1	GPIO_118	RFFE1_CLK
GPIO_37	FORCE_USB_BOOT	GPIO_78	WL_CMD_DATA_0	GPIO_119	RFFE1_DATA
GPIO_38		GPIO_79	WL_CMD_SET	GPIO_120	RFFE2_CLK
GPIO_39	MCAM_PWD_N	GPIO_80	WL_CMD_CLK	GPIO_121	RFFE2_DATA
GPIO_40	MCAM_RST_N	GPIO_81	FM_SSB1	GPIO_122	RFFE4_CLK

GPIO_123	RFFE4_DATA	GPIO_134	
GPIO_124		GPIO_135	FP_SPI_CLK
GPIO_125		GPIO_136	FP_SPI_CS
GPIO_126	RFFE3_CLK	GPIO_137	FP_SPI_MOSI
GPIO_127	RFFE3_DATA	GPIO_138	FP_SPI_MISO
GPIO_128		GPIO_139	USB_SS_SEL
GPIO_129	SCAM_RST_N	GPIO_140	FP_RST_N
GPIO_130	SCAM_PWD_N	GPIO_141	NFC_ESE_PWR_REQ
GPIO_131			
GPIO_132			
GPIO_133	SDCARD_DET_N		

PMI8952 GPIO/MPP Configuration For QRD8953

GPIO_1		MPP_1	
GPIO_2		MPP_2	GREEN_LED
		MPP_3	
		MPP_4	FLASH_STROBE_NOW

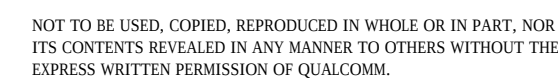
PM8953 GPIO/MPP Configuration For QRD8953

GPIO_1	CODEC_DIV_CLK	MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	NFC_CLK_REQ(SDCARD_DET_N)	MPP_2	PA_THERM1
GPIO_3	UIM_BATT_ALARM	MPP_3	VREF_DAC_MPP_3
GPIO_4		MPP_4	QUIET_THERM
GPIO_5			
GPIO_6			
GPIO_7	VBUS_USB_IN		
GPIO_8	USB_ID		

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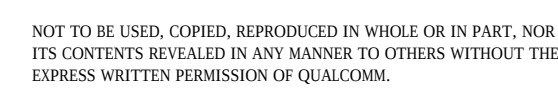
GPIO TABLE

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Size	None
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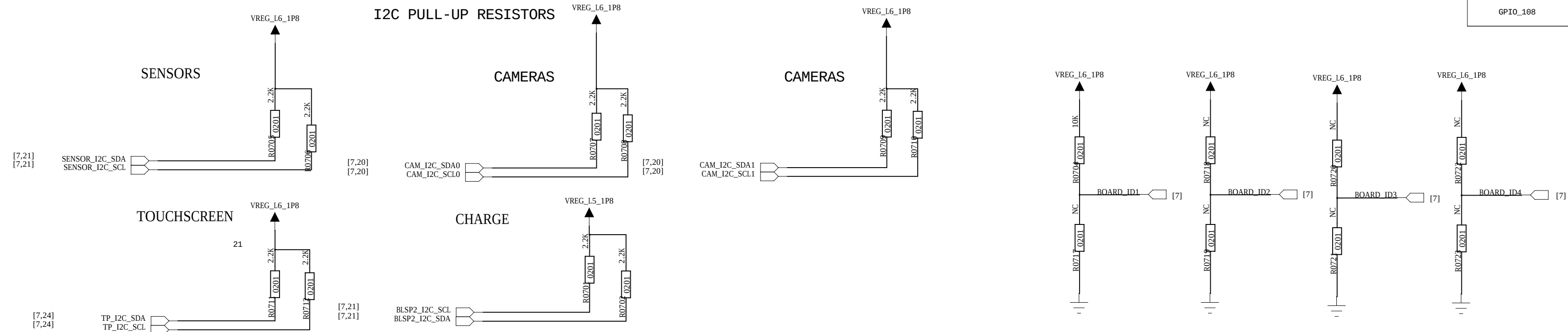
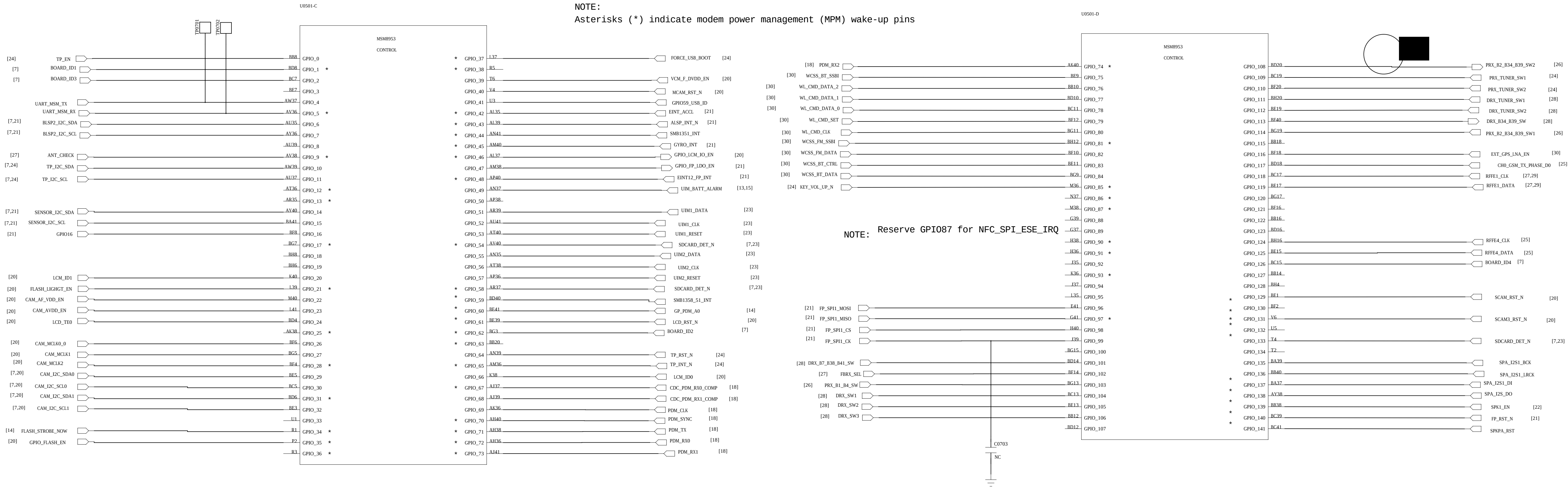
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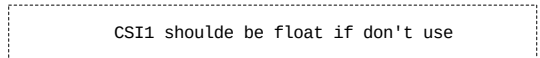
GPIO_37	FORCE_USB_BOOT
GPIO_146	MODE_DISABLE
GPIO_148	APPS_BOOT_FROM_ROM

BOOT_CONFIG[3:1]	BOOT_CONFIG
0000	SDC1 -> SDC2 -> USB2.0
0001	SDC2 -> SDC1 -> USB2.0
0010	SDC1 -> USB2.0
0011	USB2.0

Default Boot Config (0000) is SDC1@9MC

MSM8953 GPIO

NOTE:
Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to_inout pull down when the peripherals are powered off to eliminate leakage.



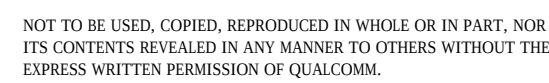
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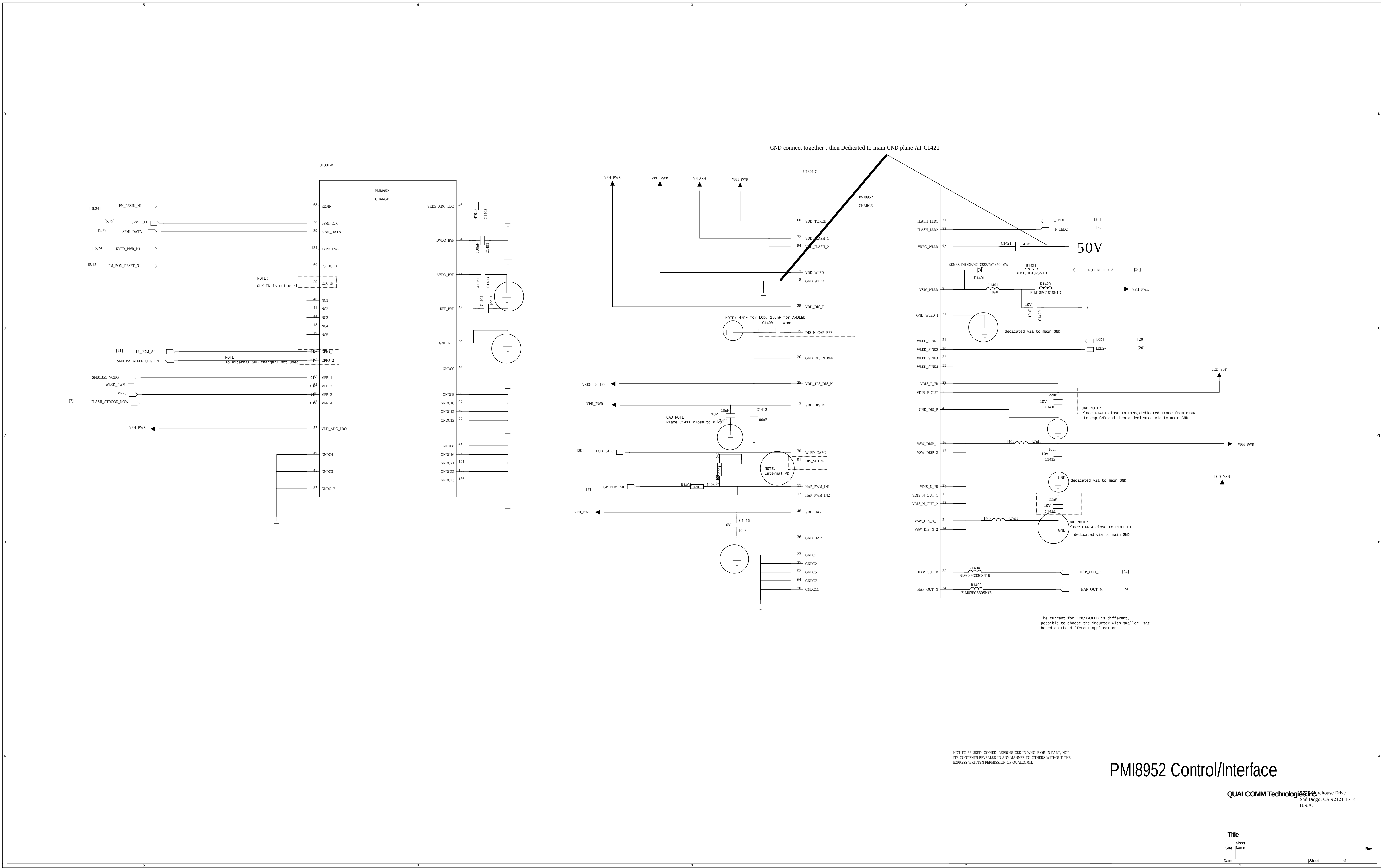
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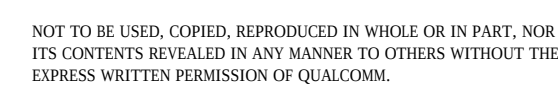
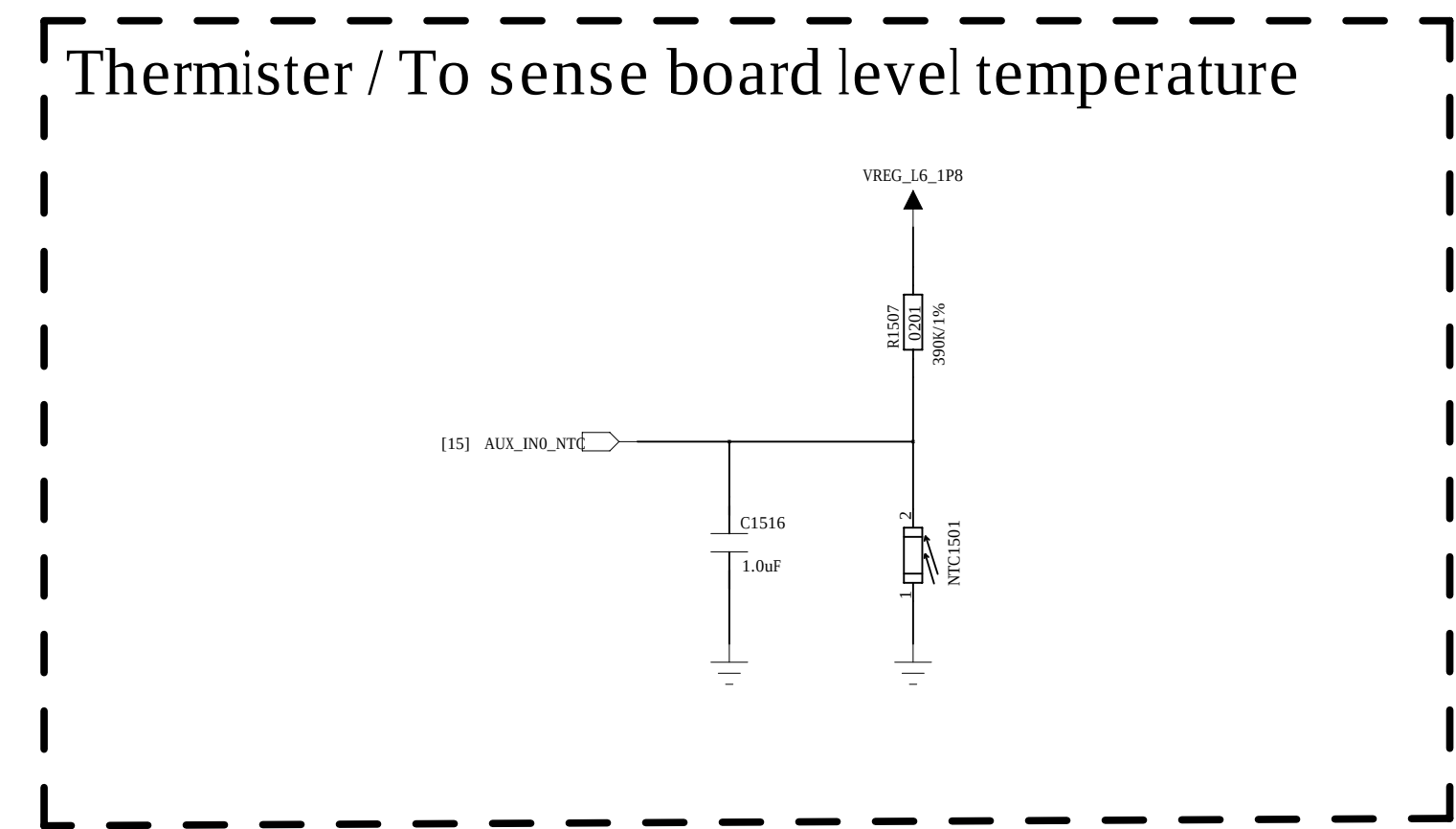


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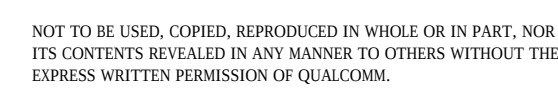
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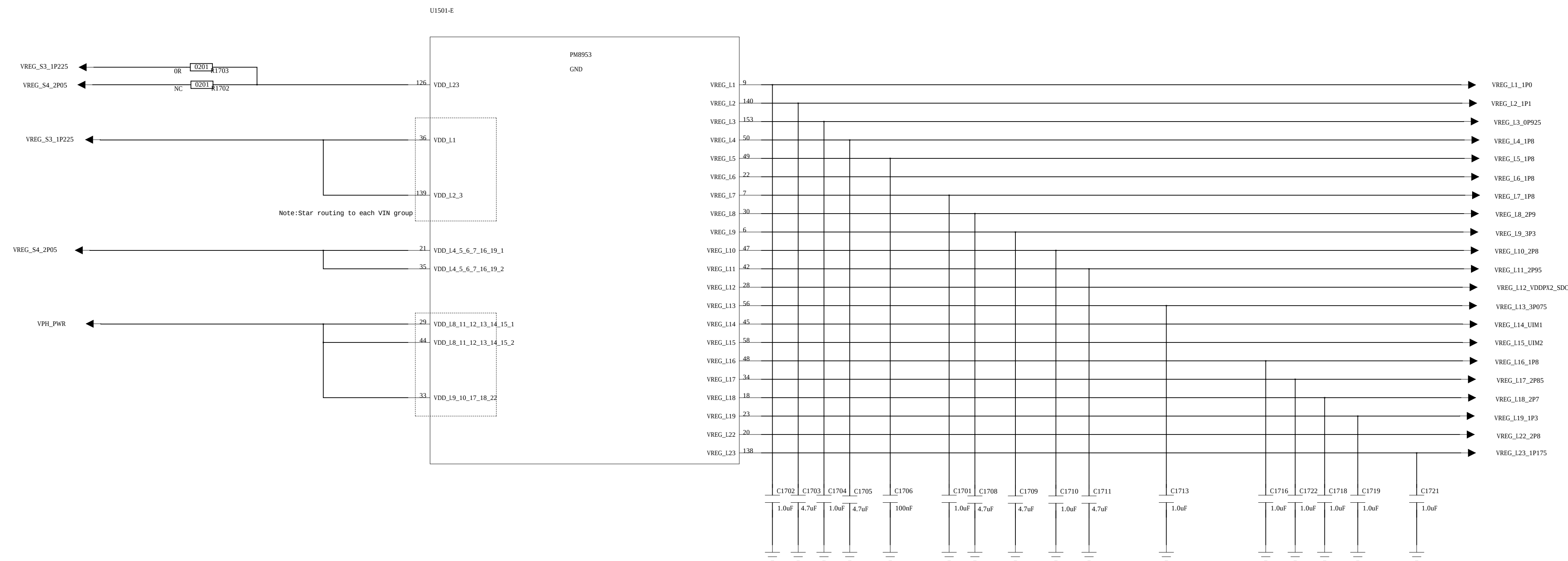


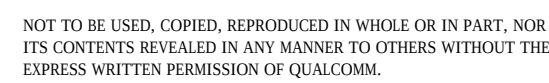
Table 3-6 PM8937/PM8940 regulators and their intended uses

Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	Expected
S1	ULT-SMPS	1.225	0.900–1.350	0.375–1.5625	2000 ³	N	MSM modem
S2	ULT-SMPS	1.225	0.550–1.350	0.375–1.5625	3000	Y	MSM core and graphics
S3	HF-SMPS	1.288	1.200–1.4125	0.375–1.5625	2700	Y	Low-voltage LDOs (1, 2, 3)
S4	ULT-SMPS	2.050	1.800–2.050	1.550–3.126	2500	Y	High-voltage LDOs (4, 5)
S5	FT-SMPS	1.225	1.050–1.350	0.350–1.355	3000	Y	MSM applications processor
S6	FT-SMPS	1.225	1.050–1.350	0.350–1.355	3000	N	MSM applications processor
L1	NMOS LDO	1.000	1.000	0.375–1.5375	1200	N	RFICs
L2	NMOS LDO	1.200	1.200	0.375–1.5375	1200	Y	LPDDR2/LPDDR3, MIP
L3	NMOS LDO	1.225	0.750–1.350	0.375–1.5375	1200	Y	VDDMX
L4	PMOS LDO	1.800	1.800	1.750–3.3375	450	N	RFICs and GPS eLNA
L5 ⁴	PMOS LDO	1.800	1.800	1.750–3.3375	500	Y	Most digital I/Os, MSM and eMMC
L6	PMOS LDO	1.800	1.800	1.750–3.3375	300	N	MSM DSI PLL and OTG display, and sensors
L7	PMOS LDO	1.800	1.800	1.750–3.3375	150	Y	MSM analog and PLL clock driver
L8	PMOS LDO	2.900	2.900	1.750–3.3375	600	Y	eMMC
L9	PMOS LDO	$V_{out} = 3.3\text{ V}$ for $VBAT > 3.575\text{ V}$; $V_{out} = 3\text{ V}$ for $VBAT < 3.575\text{ V}$	3.000–3.300	1.750–3.3375	600	N	WCN
L10	PMOS LDO	2.8	2.800	1.750–3.3375	150	N	Sensors
L11 ⁵	PMOS LDO	2.950	2.950	1.750–3.3375	800	Y	Micro SD
L12 ⁴	PMOS LDO	2.950	1.800/2.950	1.750–3.3375	150	Y	MSM pad group 2 and 3
L13	PMOS LDO	3.075	3.075	1.750–3.3375	50	Y	MSM USB and audio

Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	
L14 ⁵	PMOS LDO	1.800	1.800/2.925/3.050	1.750–3.3375	50	N	MSM pac
L15 ⁵	PMOS LDO	1.800	1.800/2.925/3.050	1.750–3.3375	50	N	MSM pac
L16	PMOS LDO	1.800	1.800	1.750–3.3375	5	N	PMIC HK
L17	PMOS LDO	2.850	2.850	1.750–3.3375	600	N	Camera,
L18	PMOS LDO	2.700	2.700	1.750–3.3375	150	N	QTI RF fr
L19	NMOS LDO	1.350	1.350	0.375–1.5375	1200	N	MSM ana
L20	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC XO
L21	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC RF
L22	PMOS LDO	2.800	2.800	1.750–3.3375	300	N	Camera -
L23	NMOS LDO	1.3	1.2	0.375–1.5375	300	N	Camera -

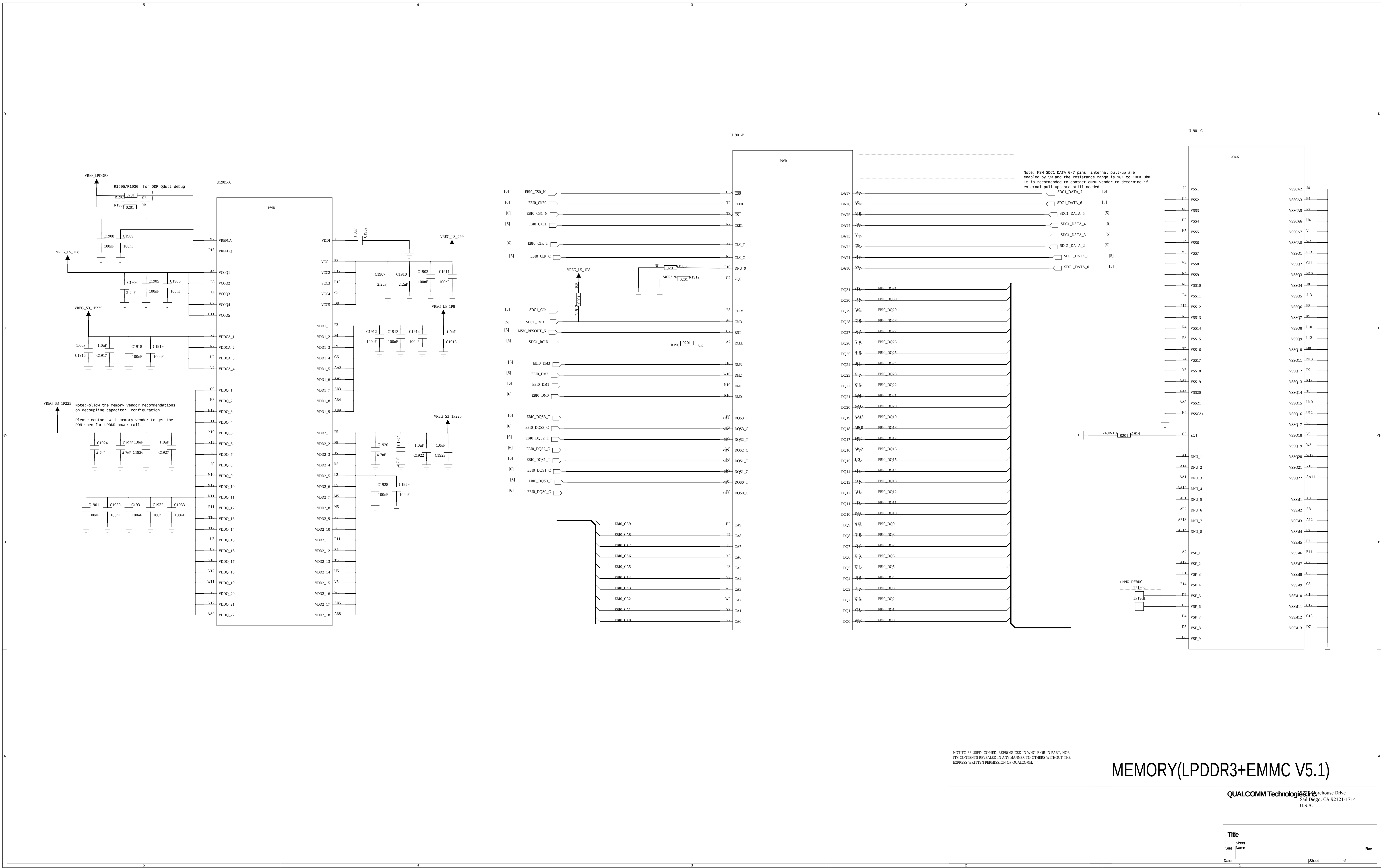
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PM8953 LDO



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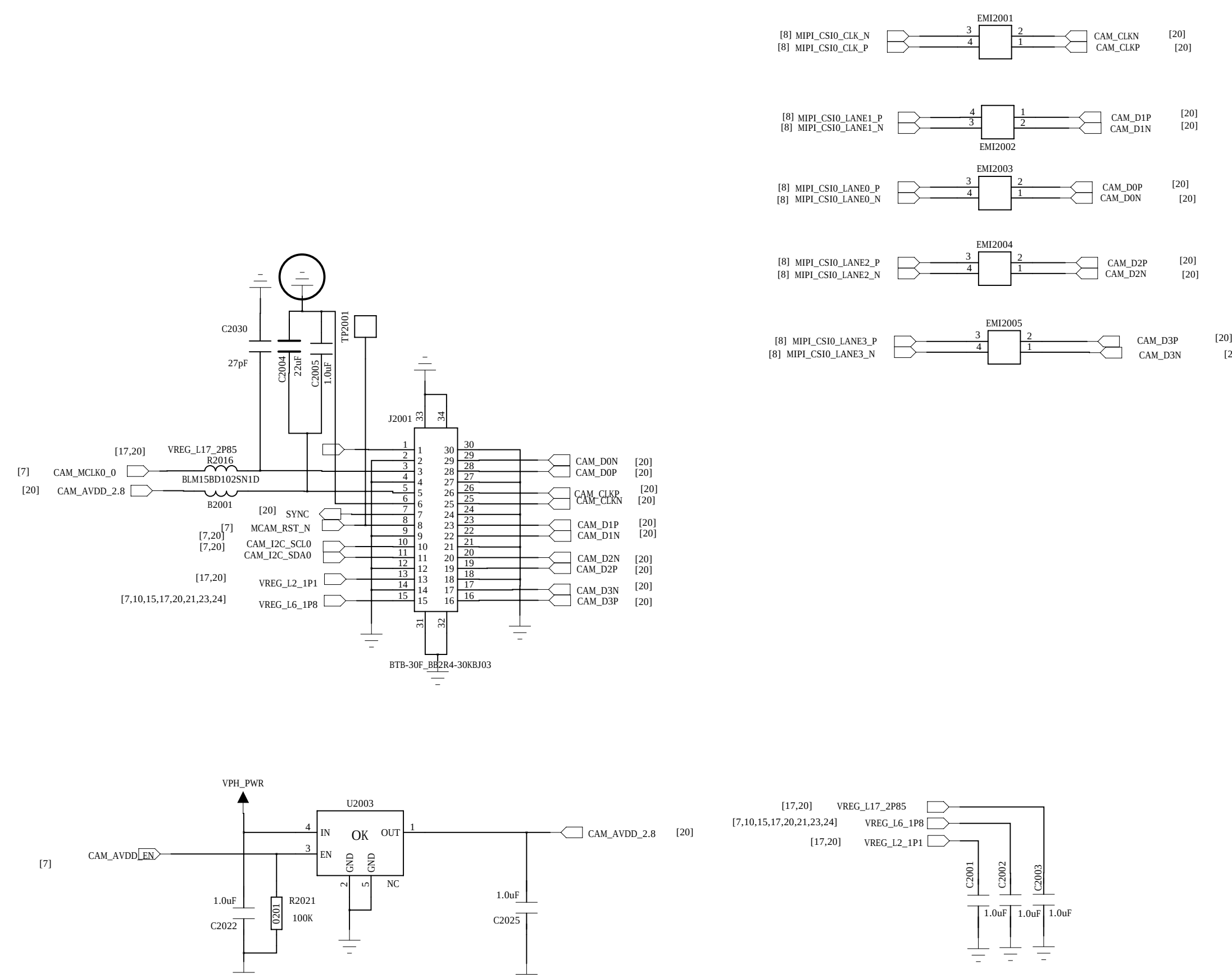
MEMORY(LPDDR3+EMMC V5.1)

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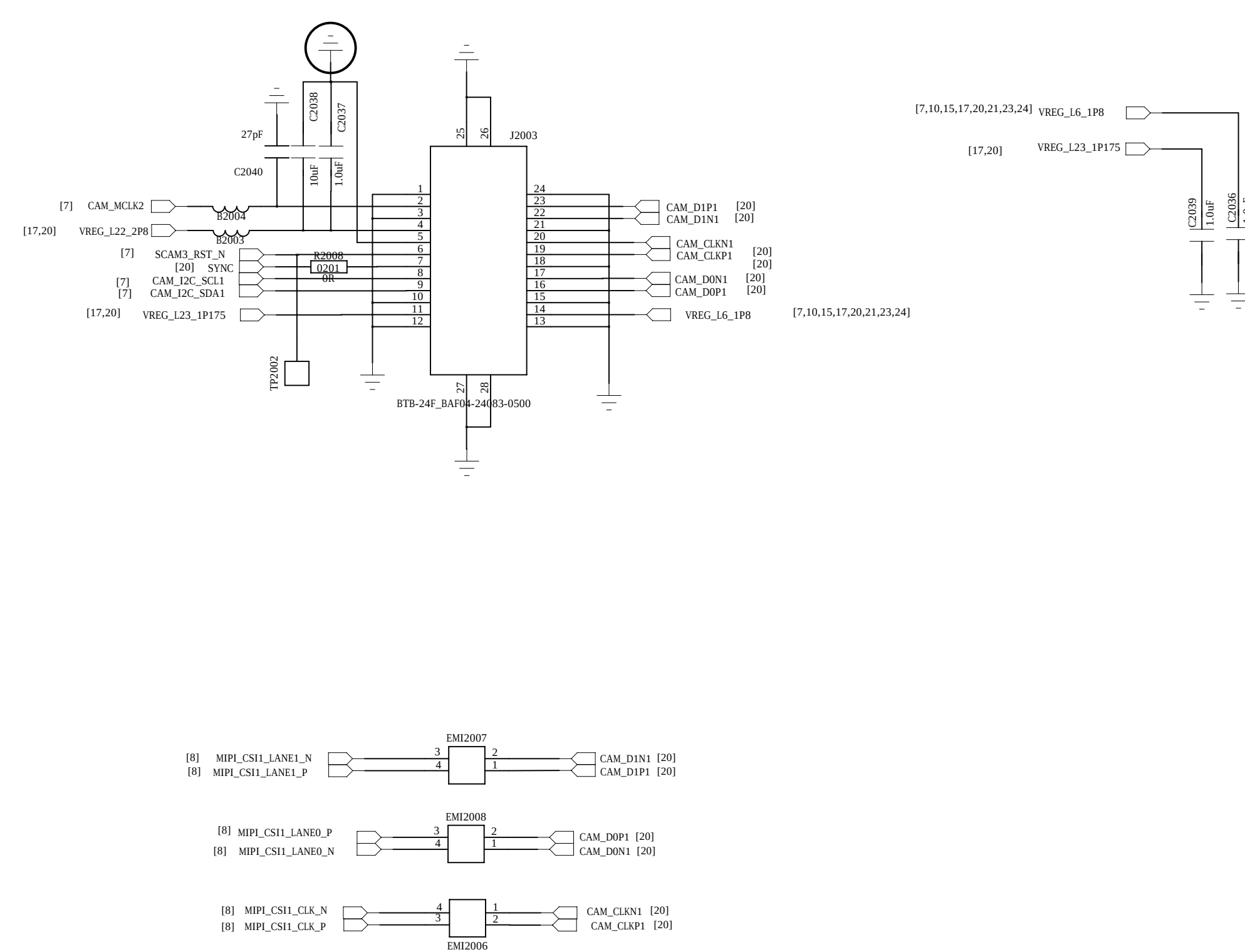
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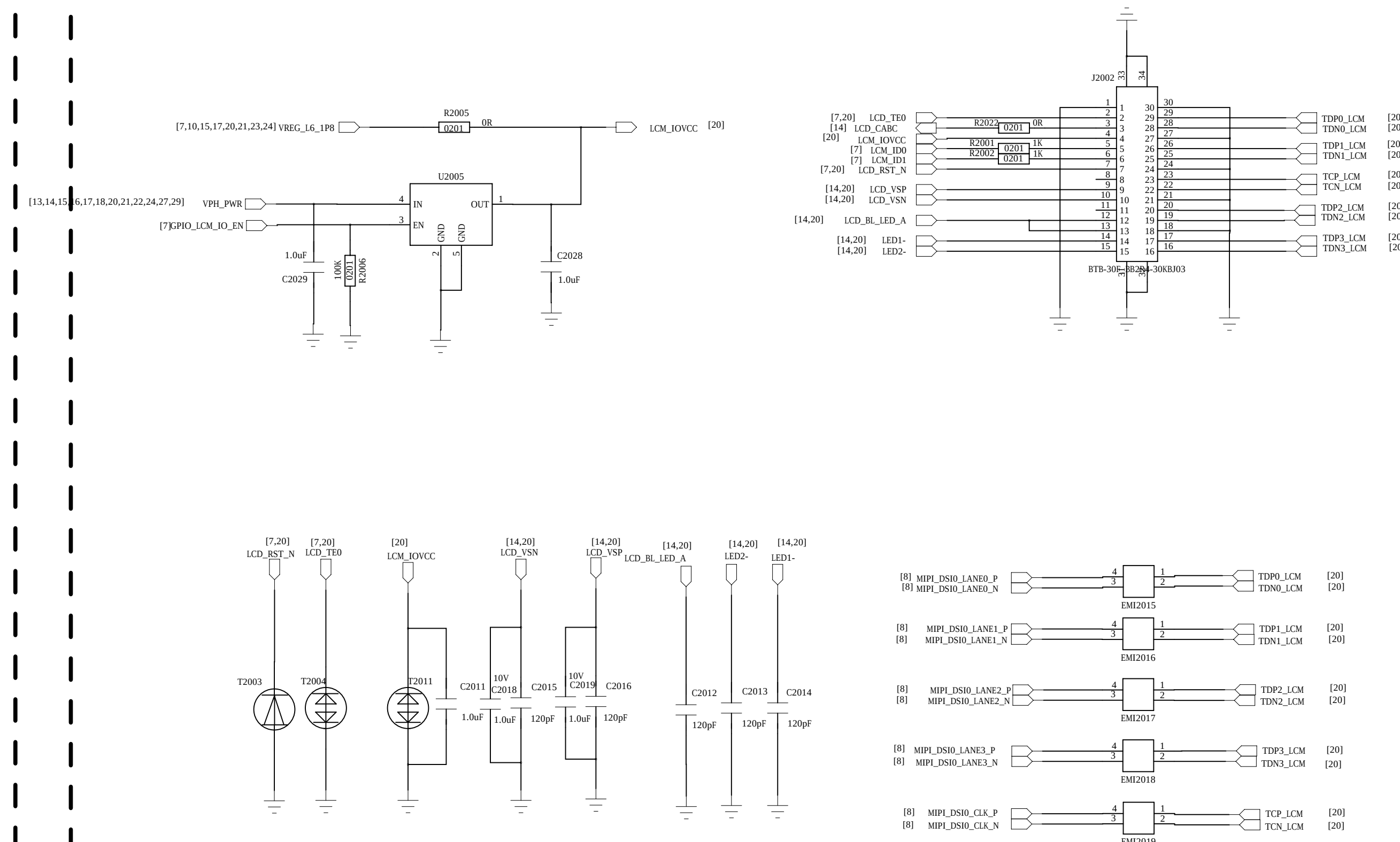
Main Camera A



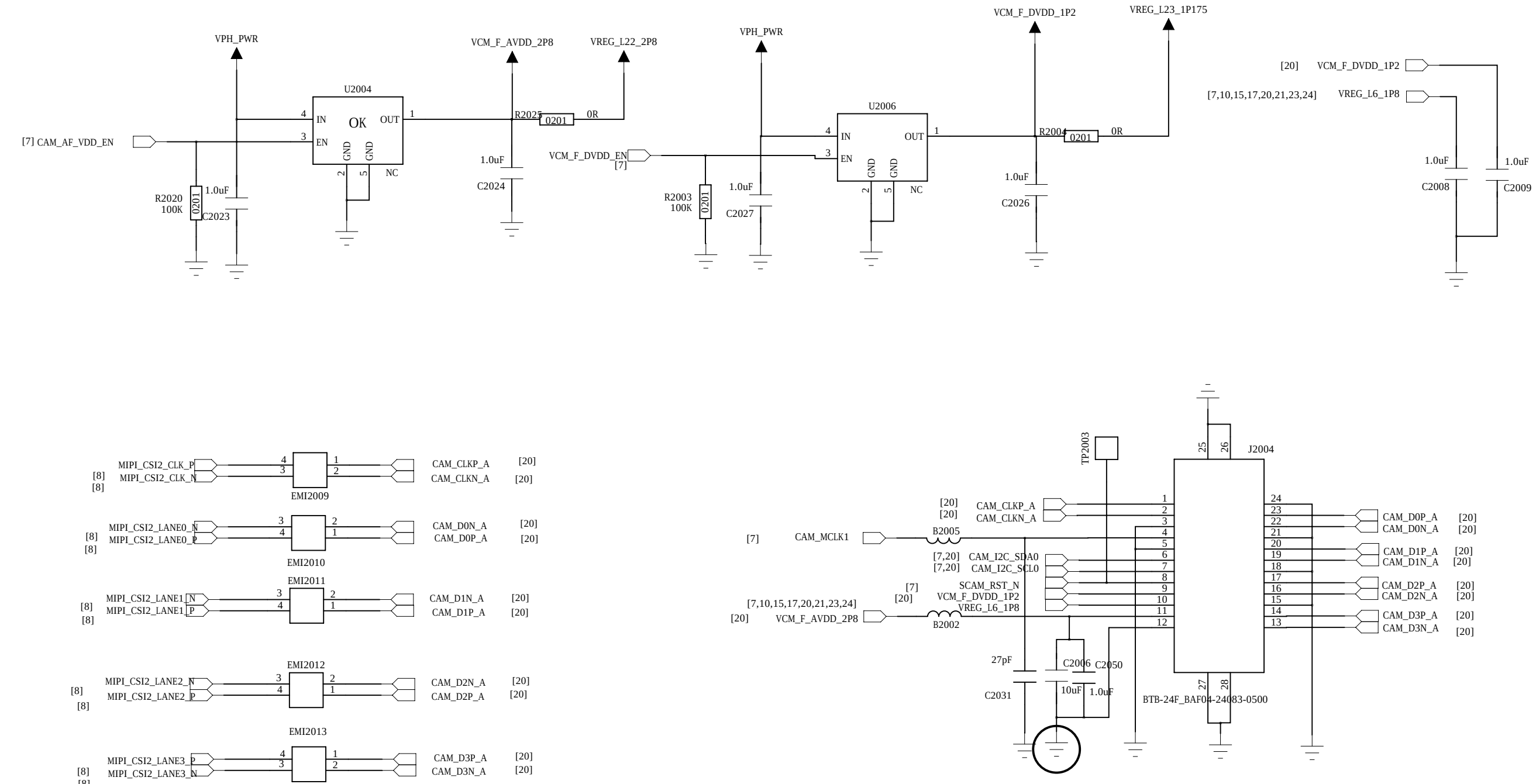
Main Camera B



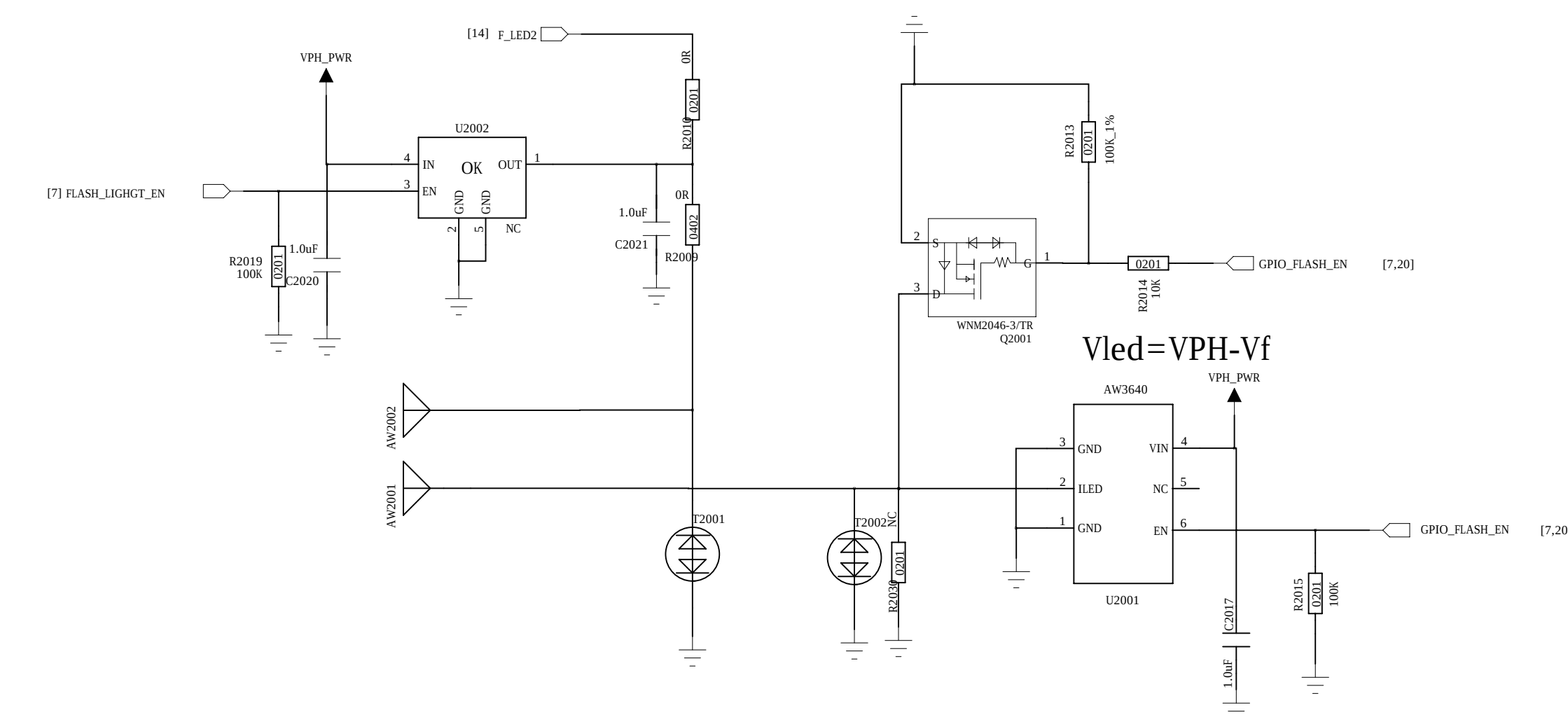
LCM



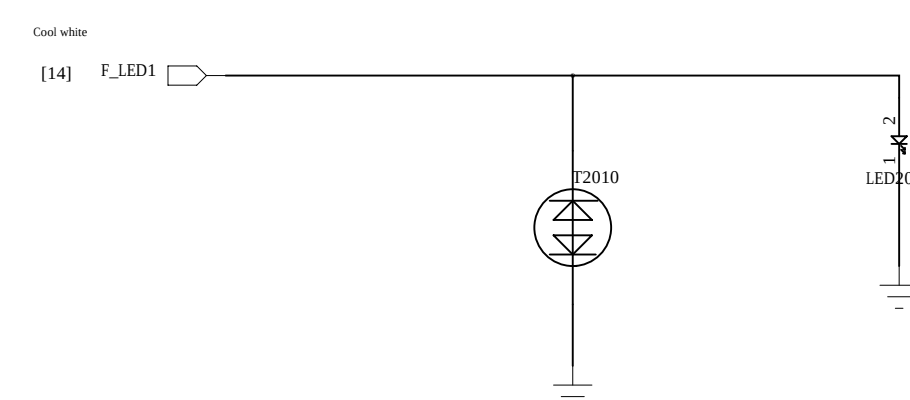
Front Camera



Front FLASHLIGHT



Main FLASHLIGHT



Schematic design notice of "63_PERI_CAMERA_KEYPAD" page.

Note 62-1: The VCC of I2C_0 is pulled to "VCAM_IO_PMU".

Note 62-2: I2C control interface of front camera (with AF) must be assigned to I2C-2 bus when PIP/VIV feature be supported.

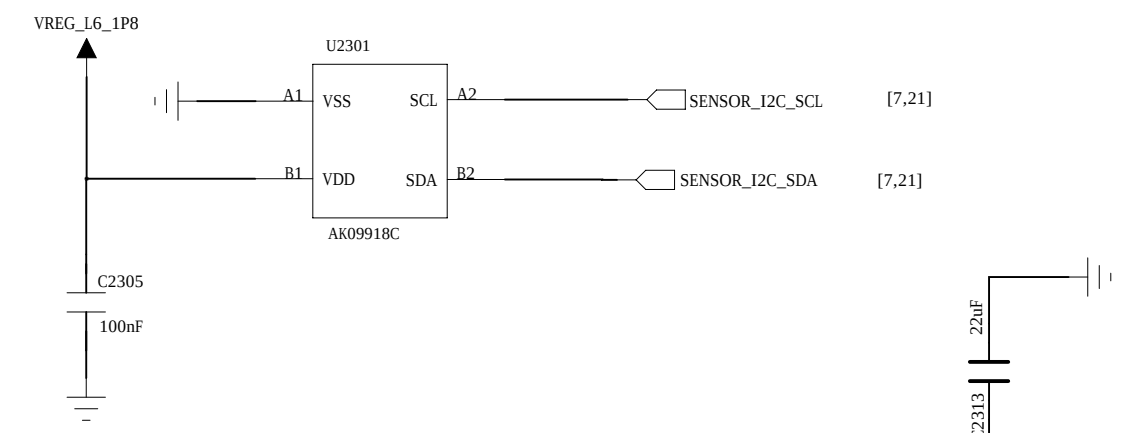
Note 62-3: Reserve a capacitor (27pF) on camera's MCLK and shunt it to GND to prevent GPS de-sense.

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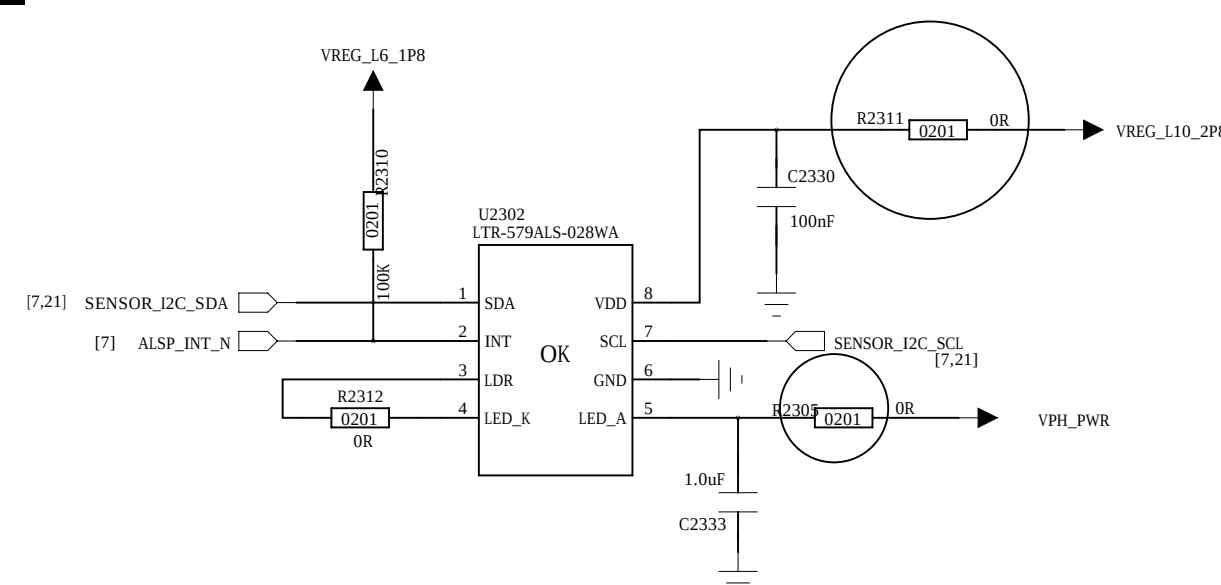
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M-Sensor

M-Sensor I2C Address: 0x0C

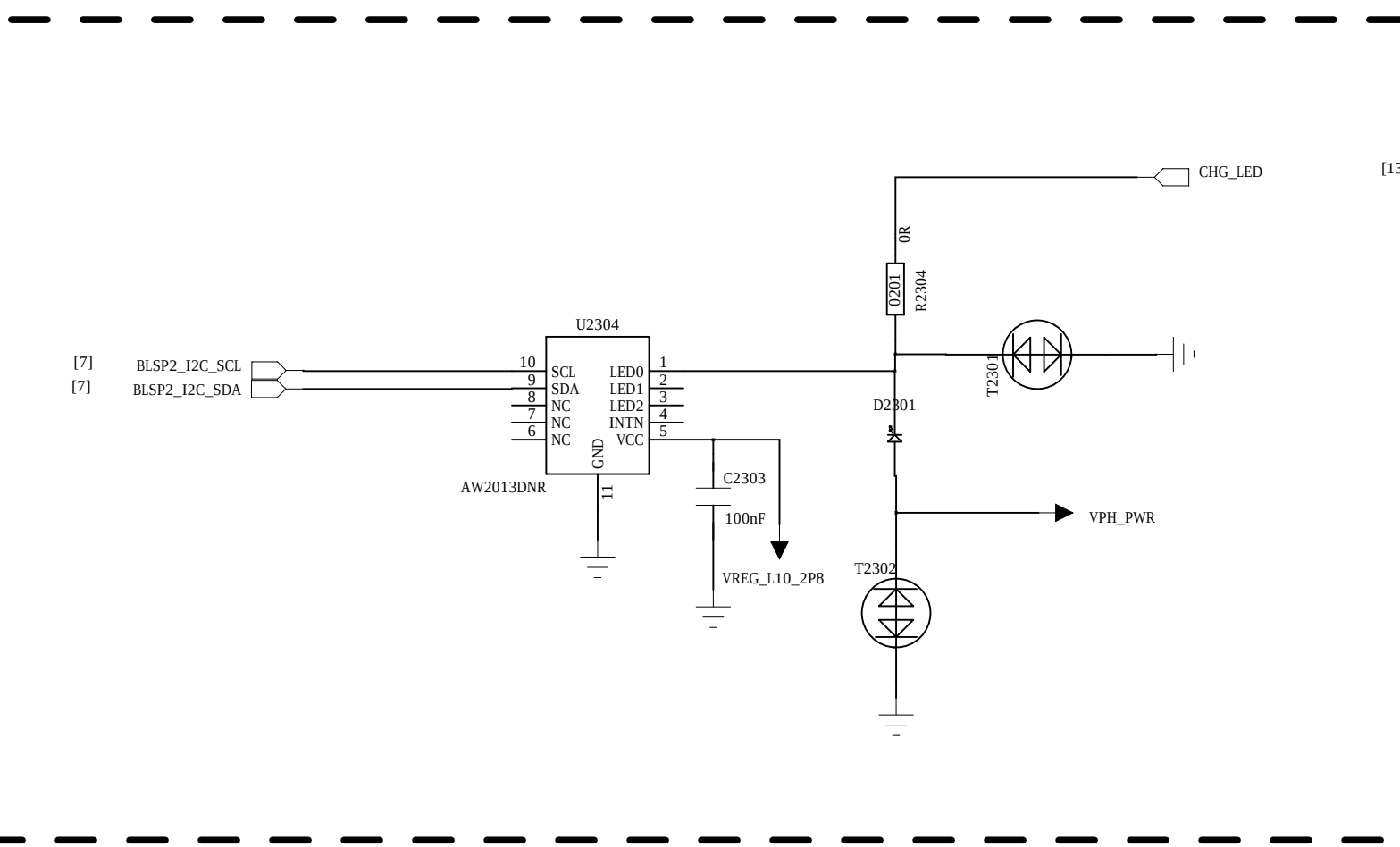
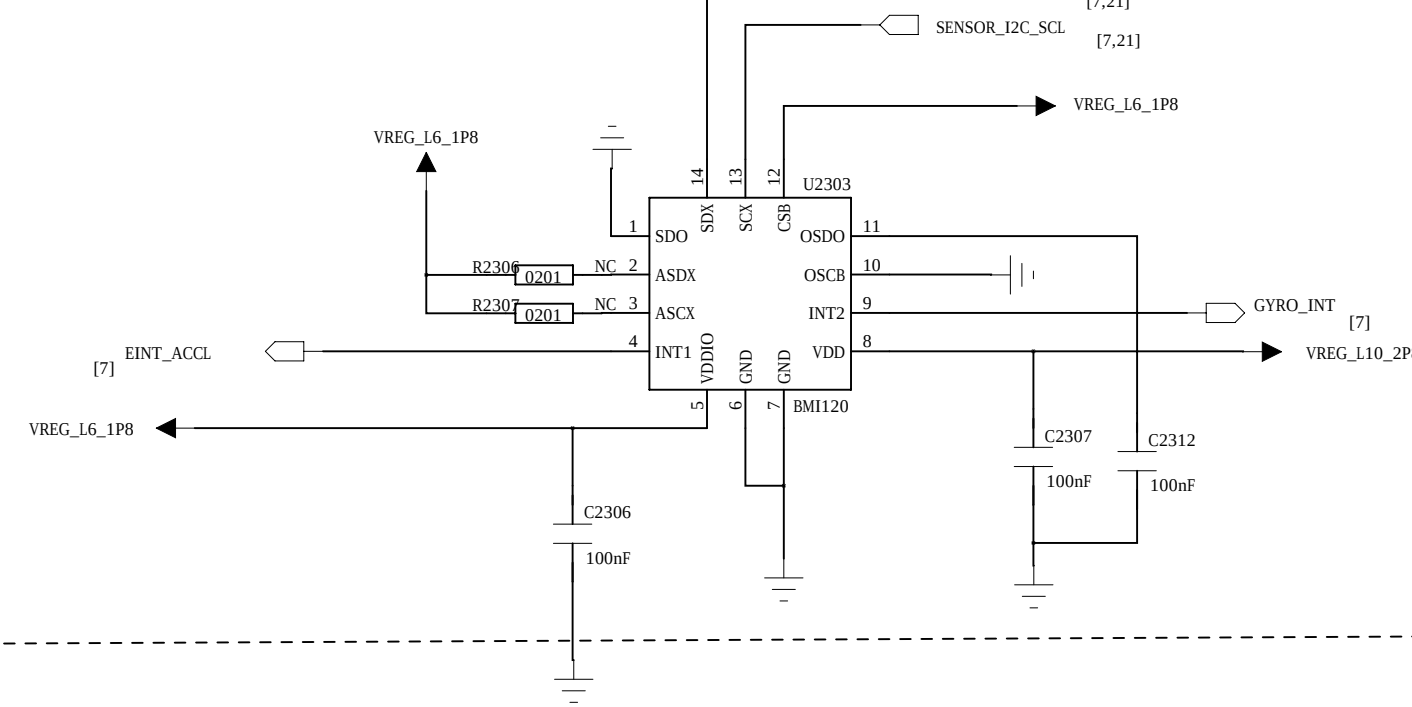


P_SENSOR

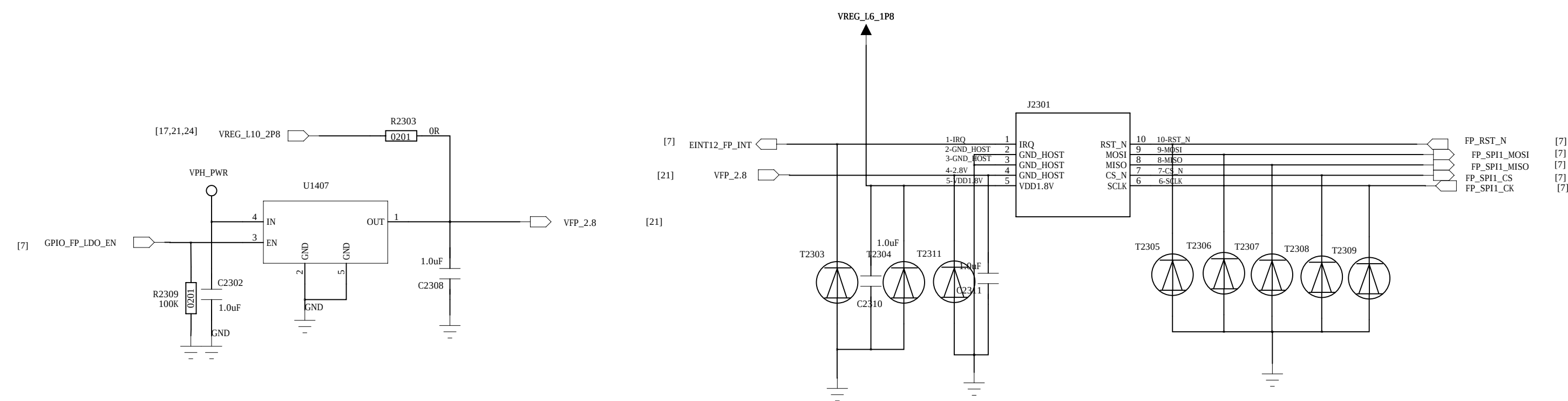


G+Gyro-Sensor

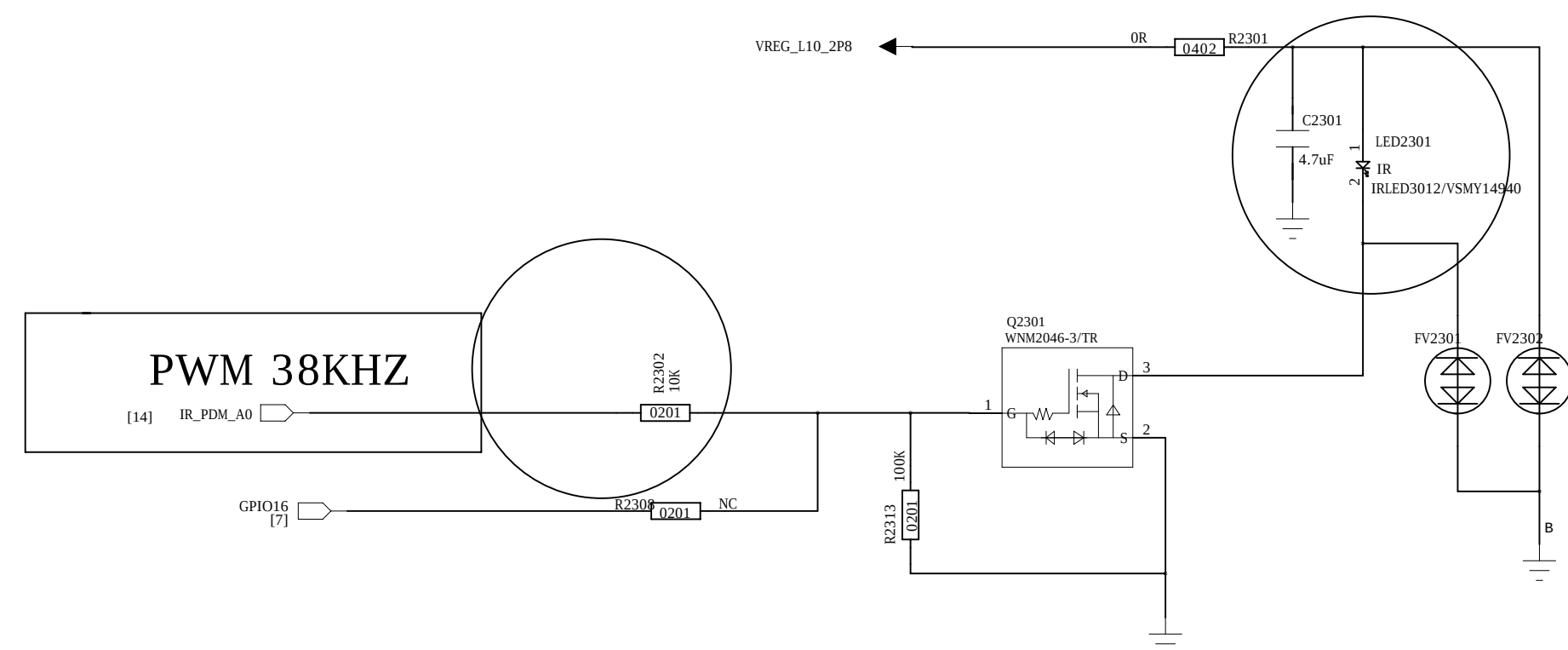
BMI160 and BMI120 is Pin To Pin I2C address : 0x68



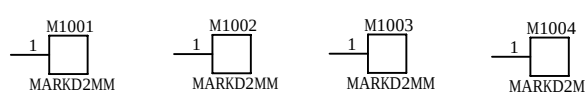
FingerPrint-Sensor



IR

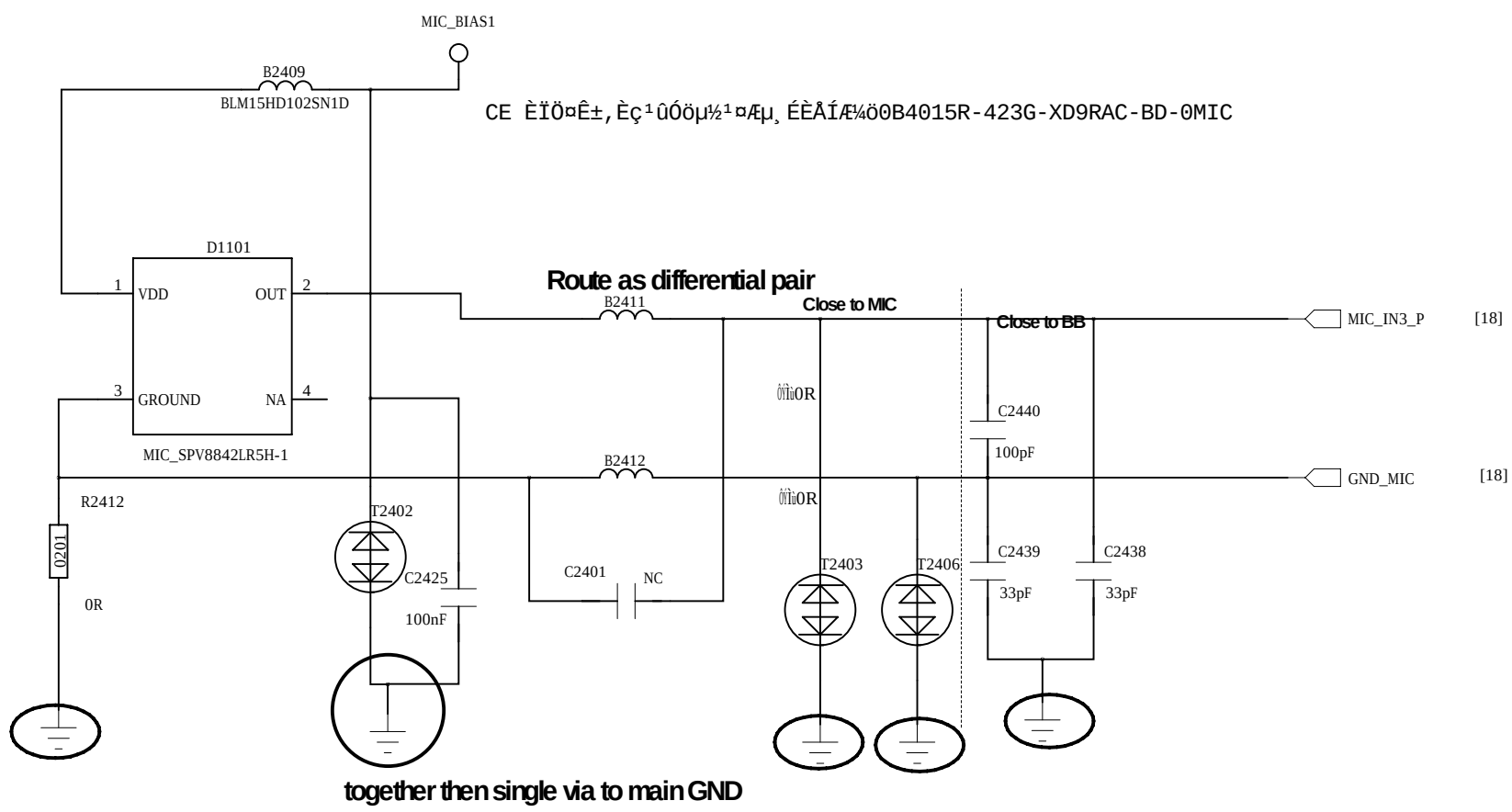


MARK POINT

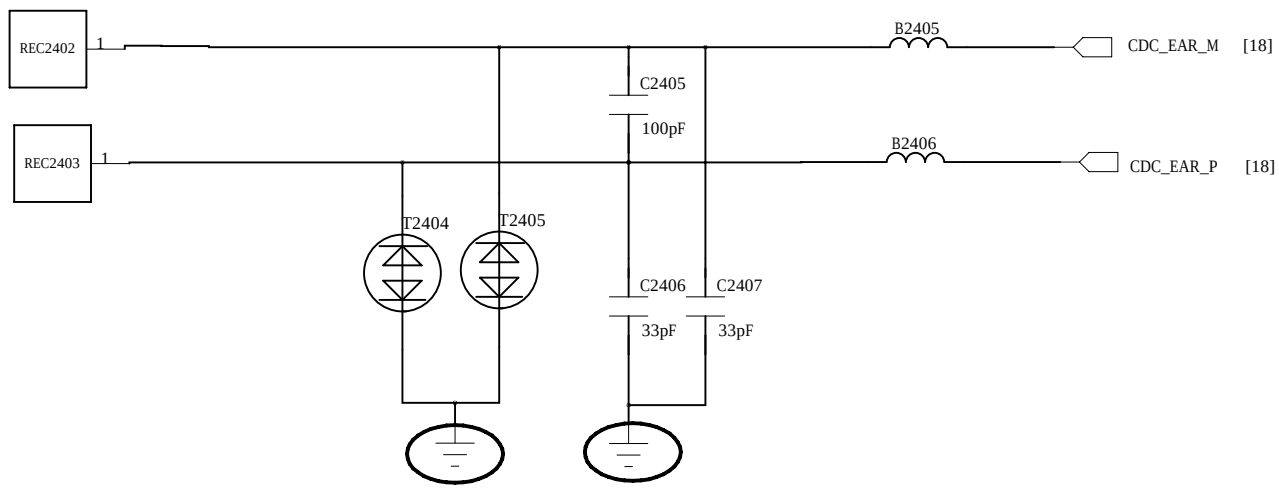


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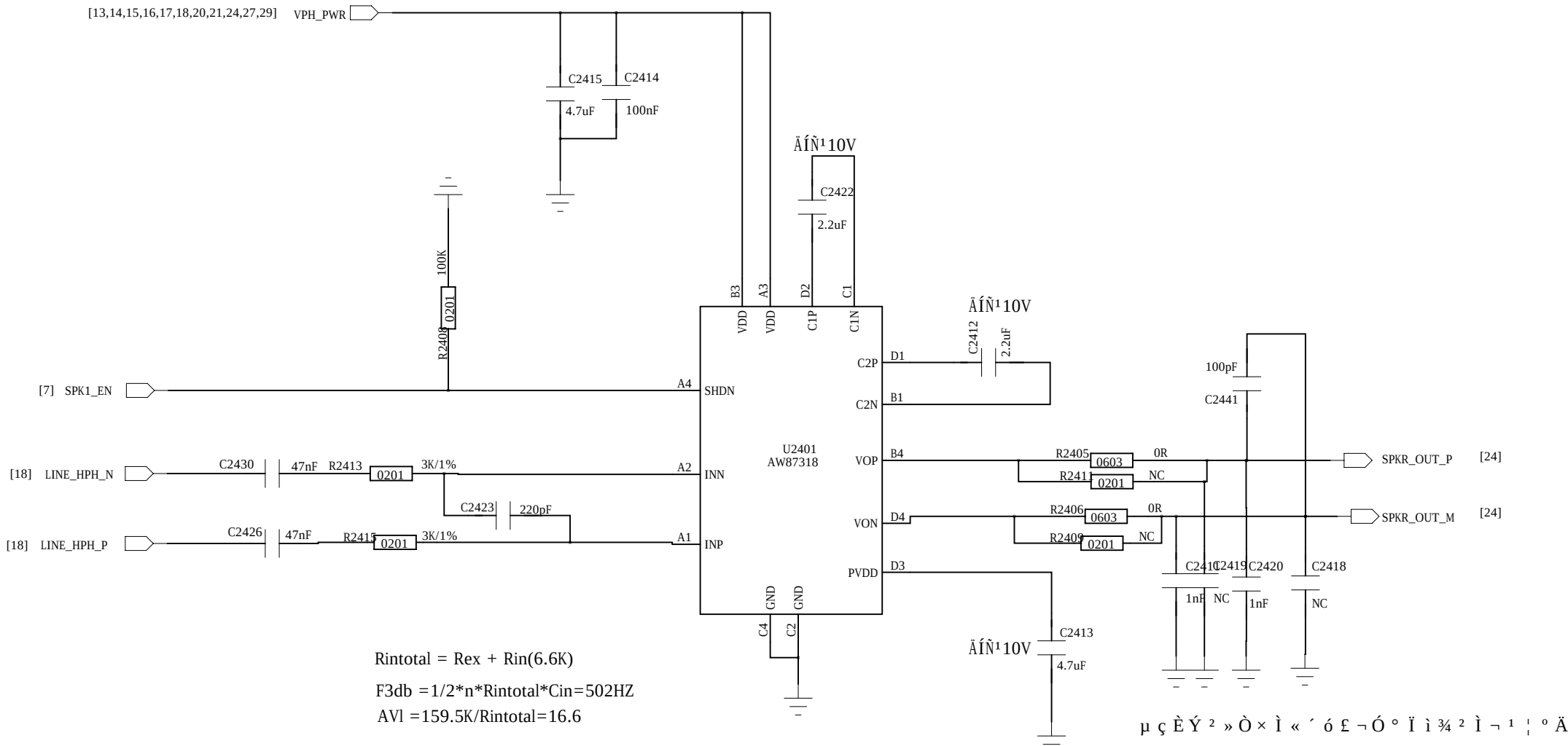
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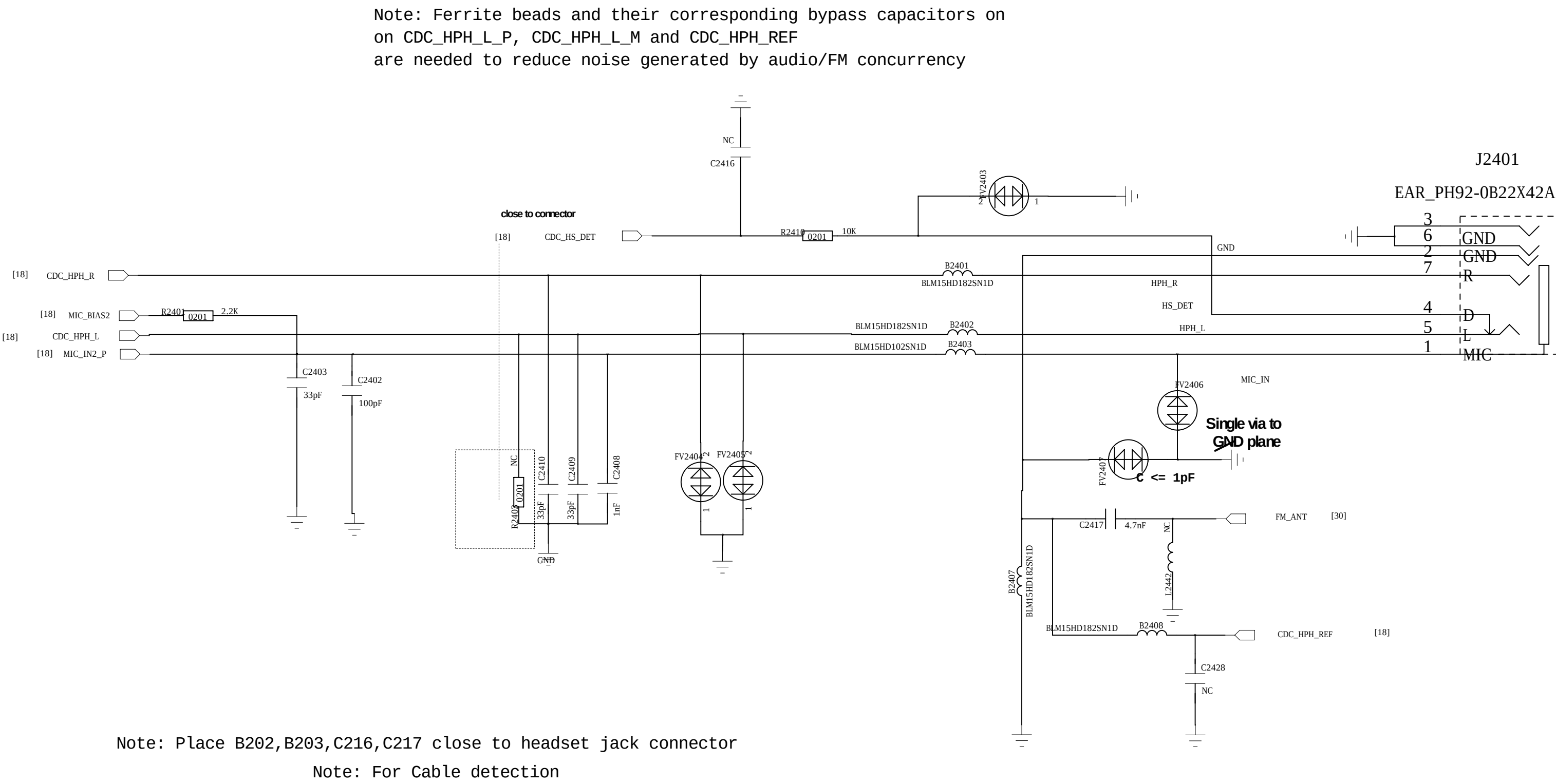
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Audio PA

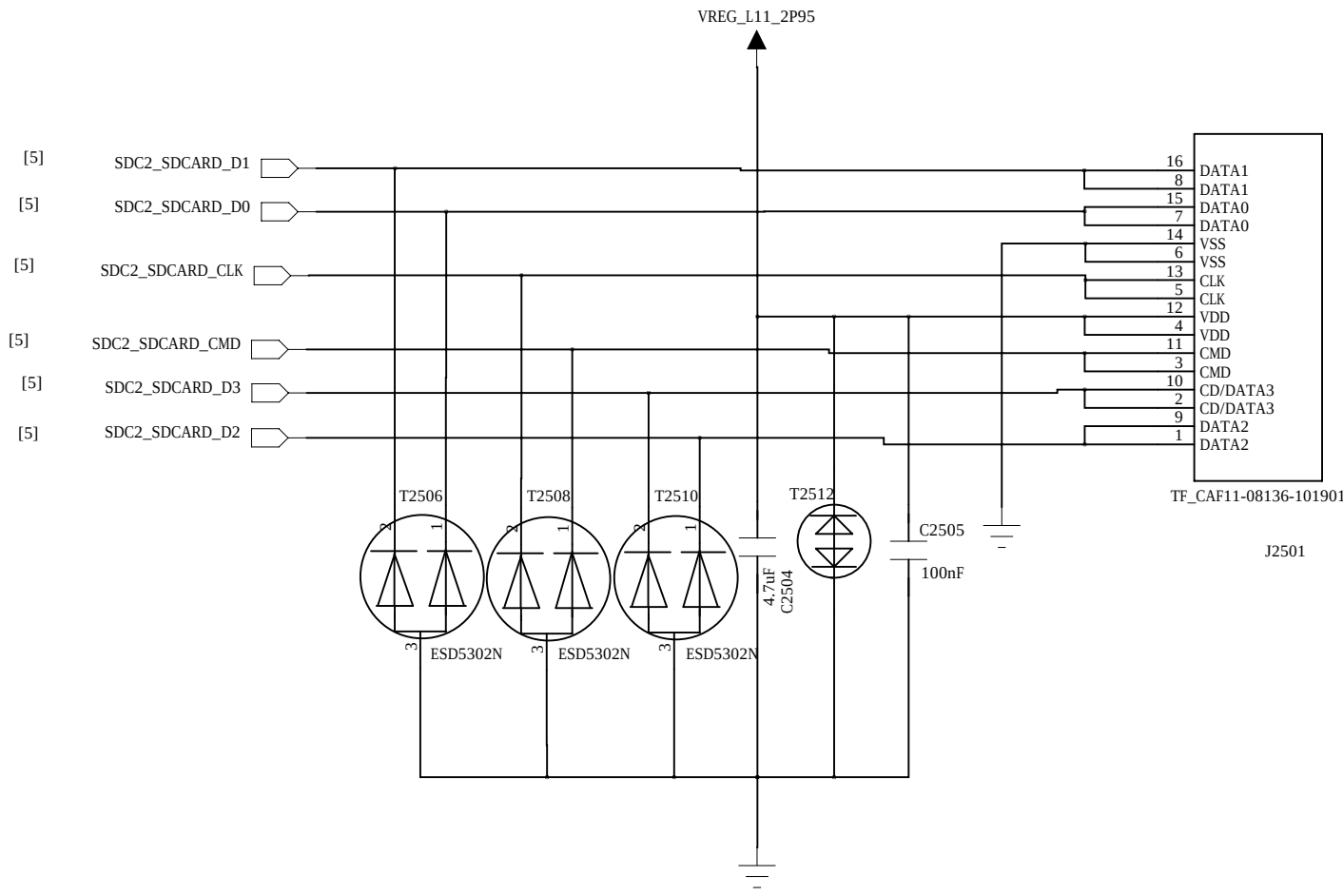


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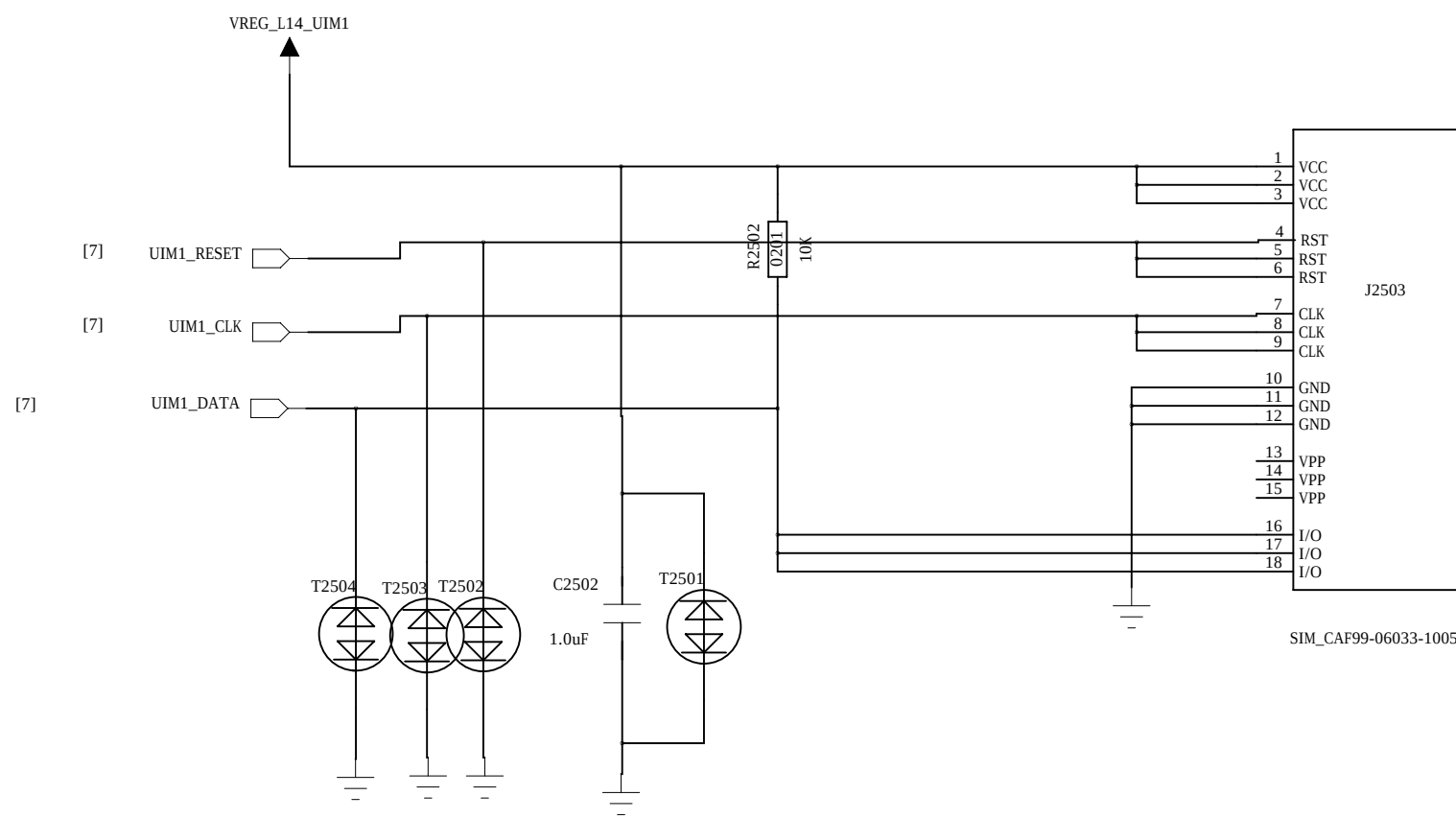


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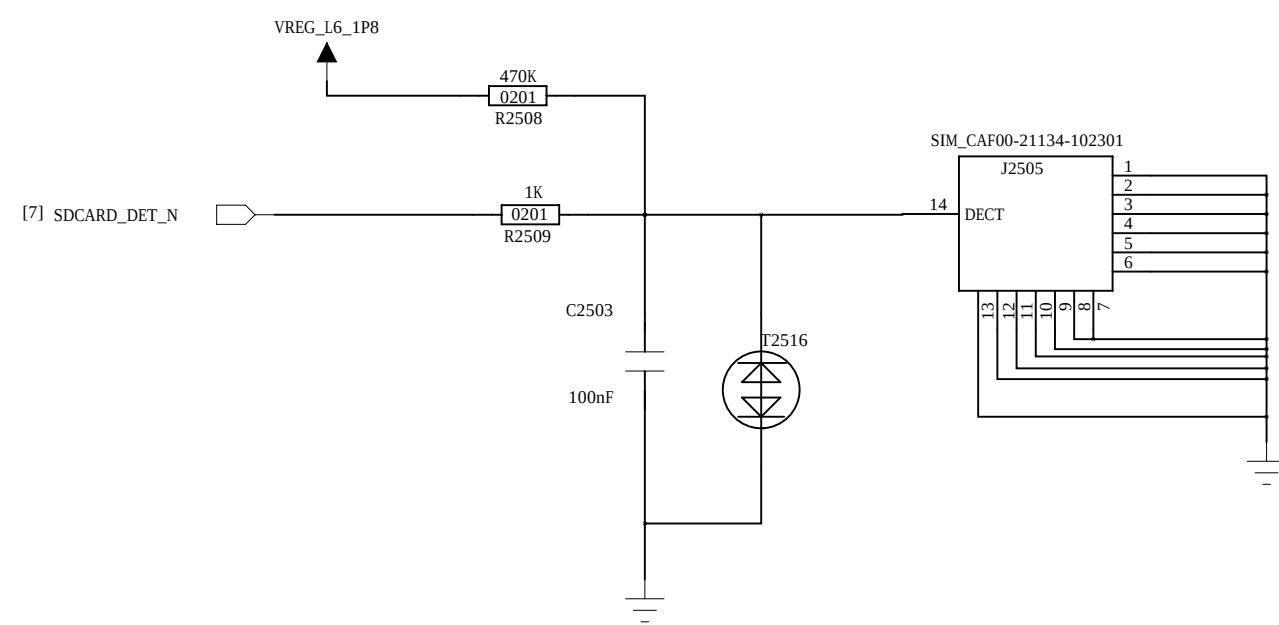
T-Card



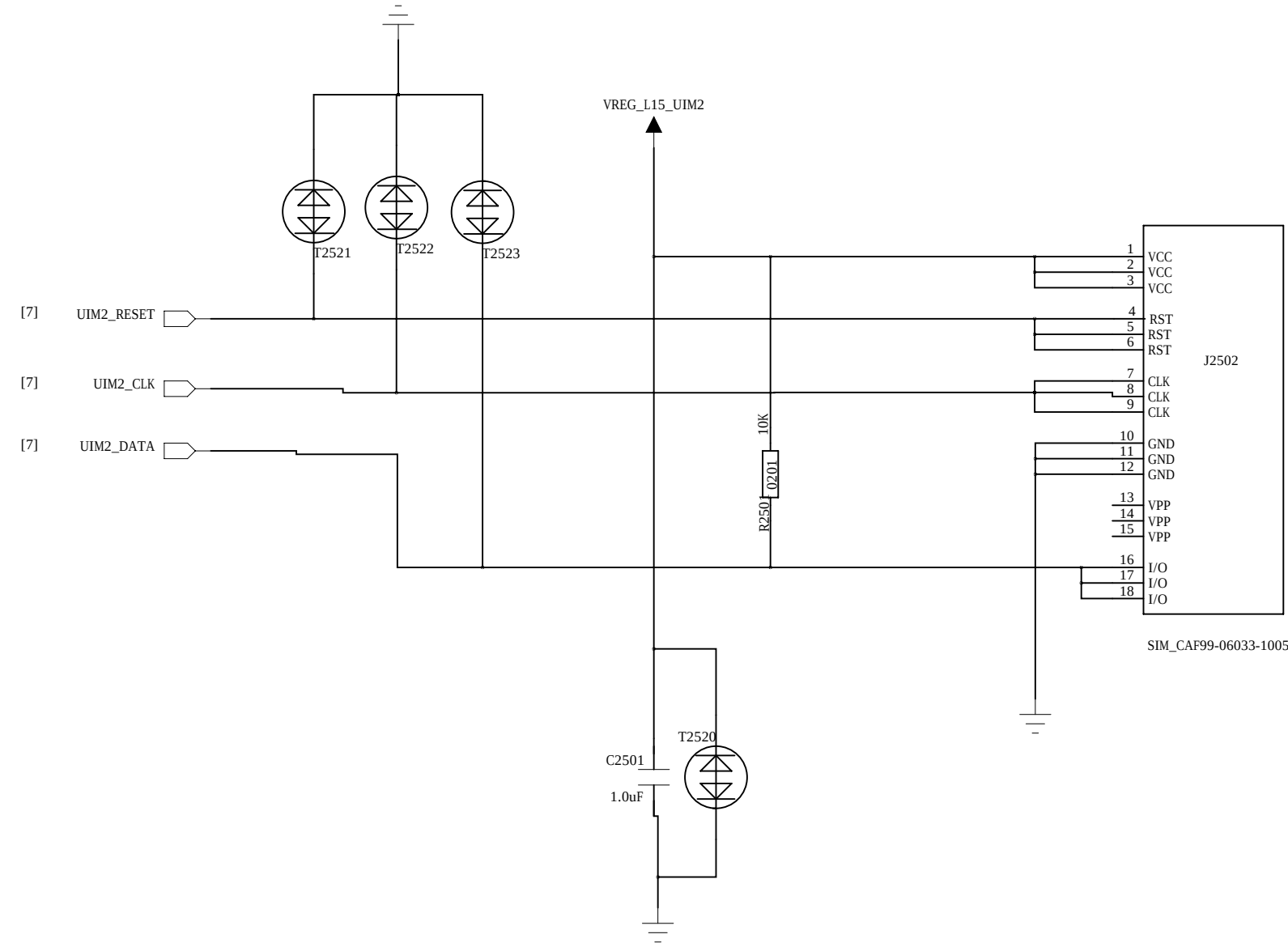
SIM1



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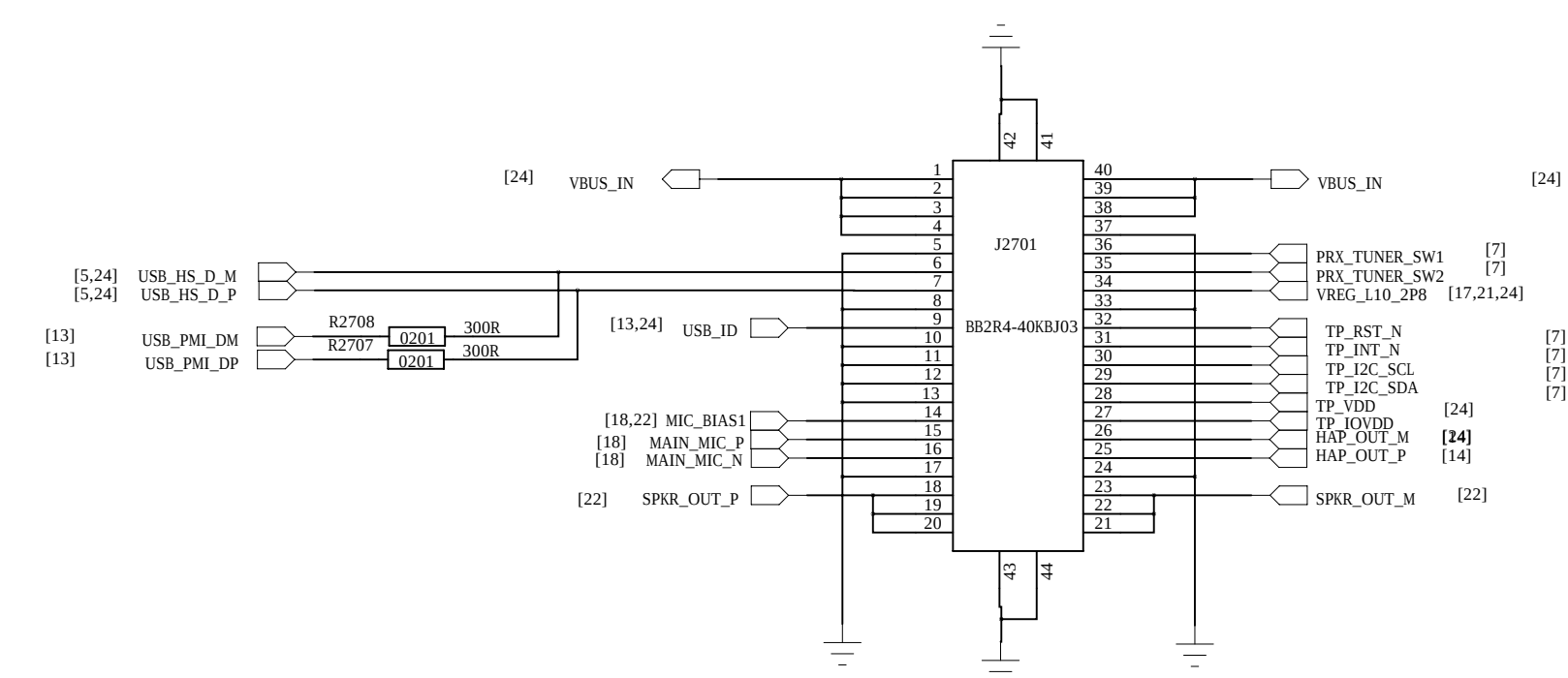


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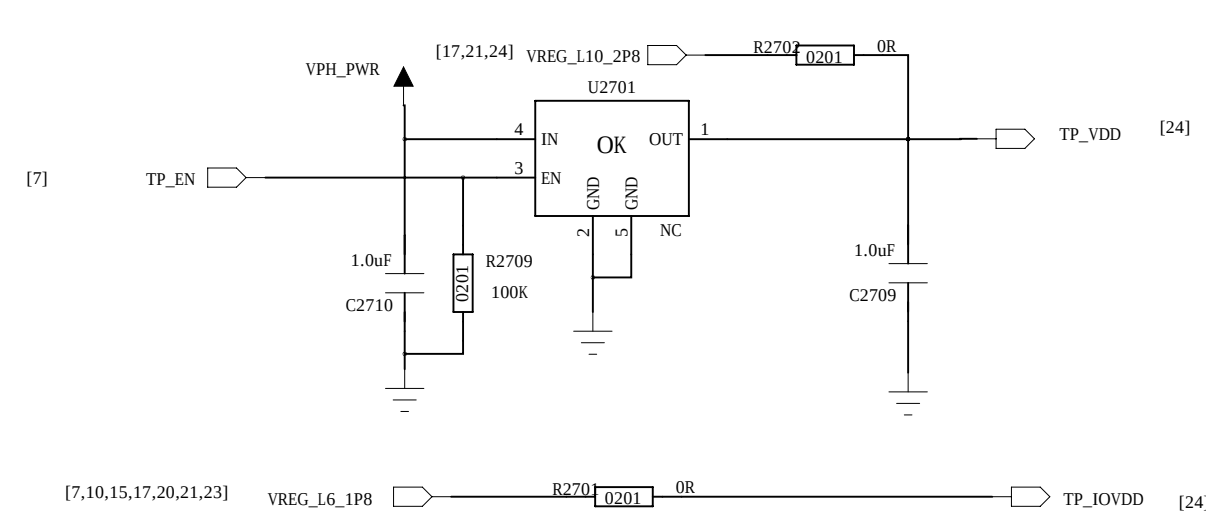


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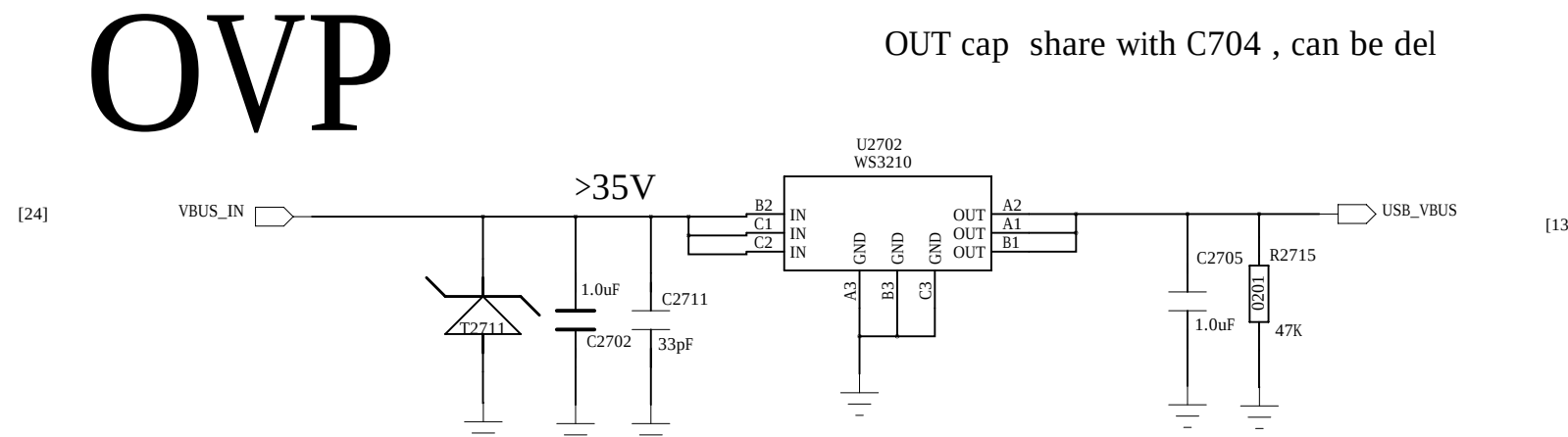
SUB_FPC



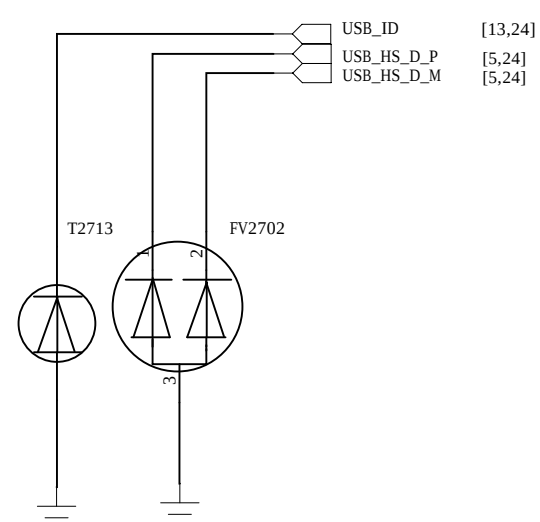
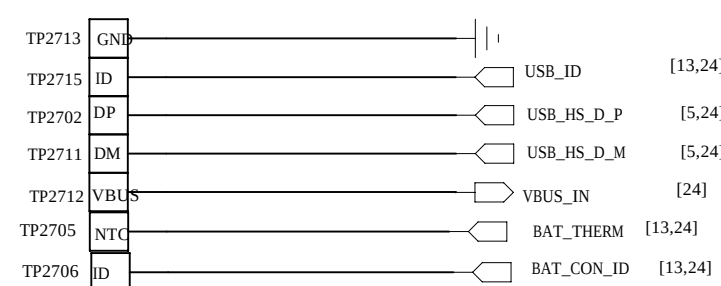
LDO



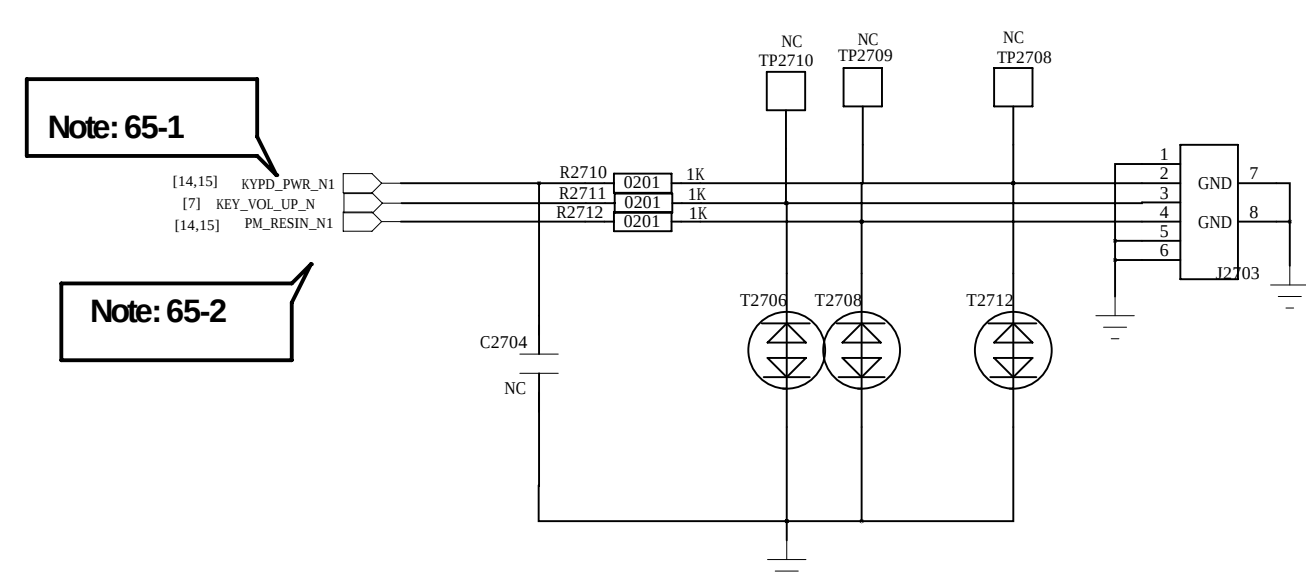
OVP



Test point



Power Key

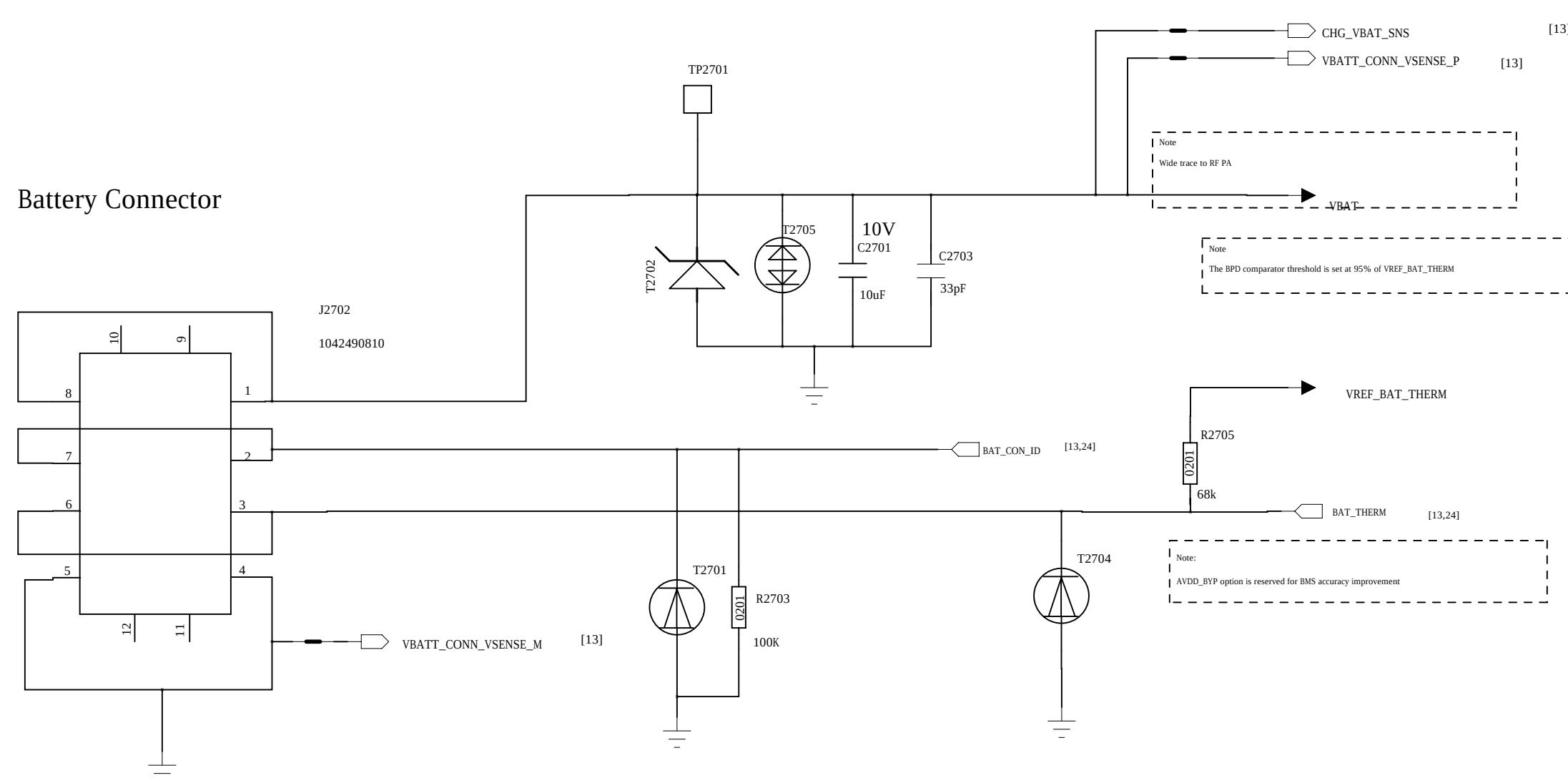


Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

Note 65-1: DO NOT put pull-up resistor on PWRKEY

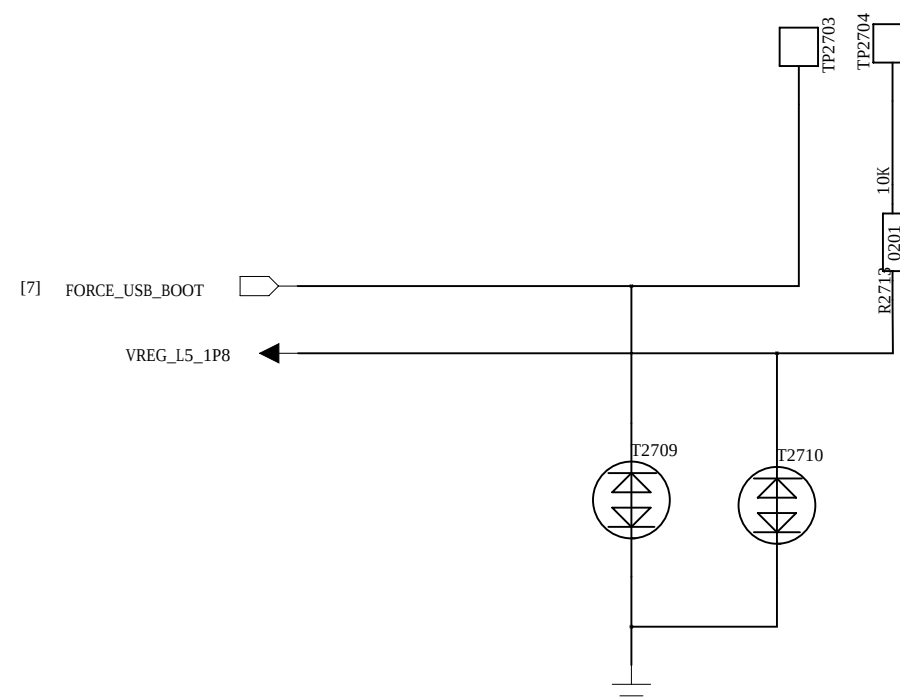
Note 65-2: Volume Up : HOME Key / GND
Volume Down : (KPROW0/KPCOL0) or KPCOL0 / GND

BAT



Note:
Battery ID resistor value require 20K~150K.
Add 100K to ground if battery package ID not meet requirement

Force USB boot



D

D

C

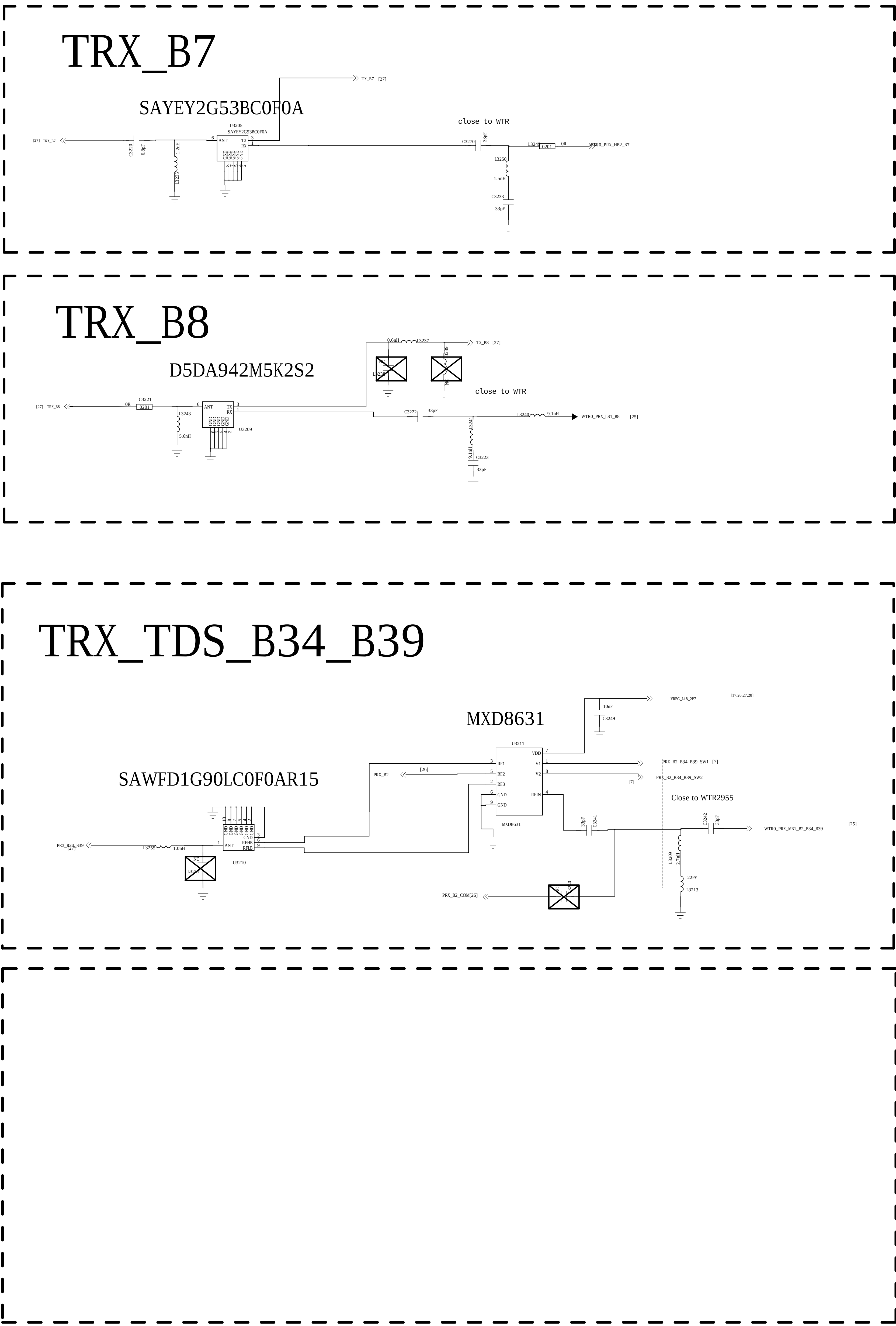
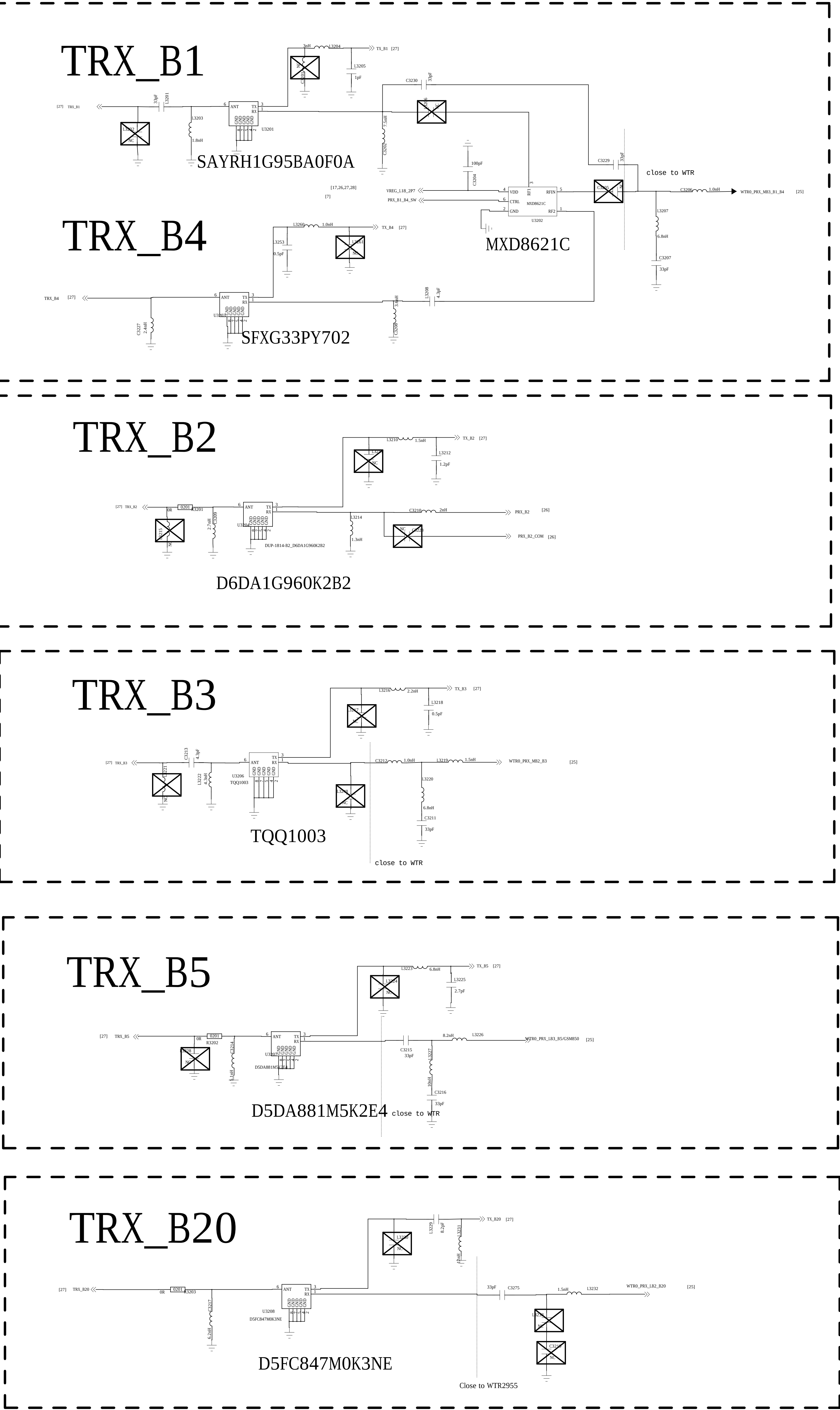
C

B

B

A

A



REVISION RECORD			
LYR	REV. NO.	APPROVED:	DATE:

TXM

X.FL-R-SMT-1(80)

MM8030-2610B

RFMD5212A

MXD8621C

MMPA

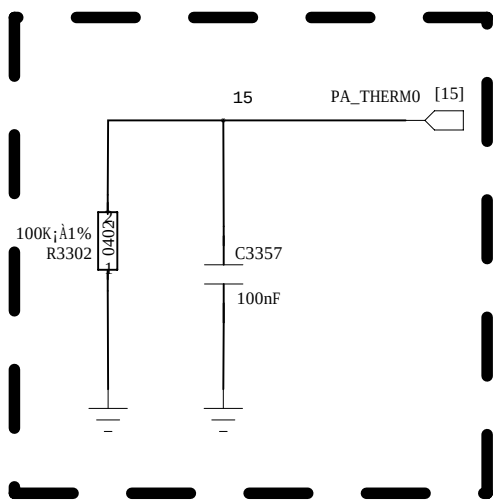
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ACX BF1608-L1R9NDB

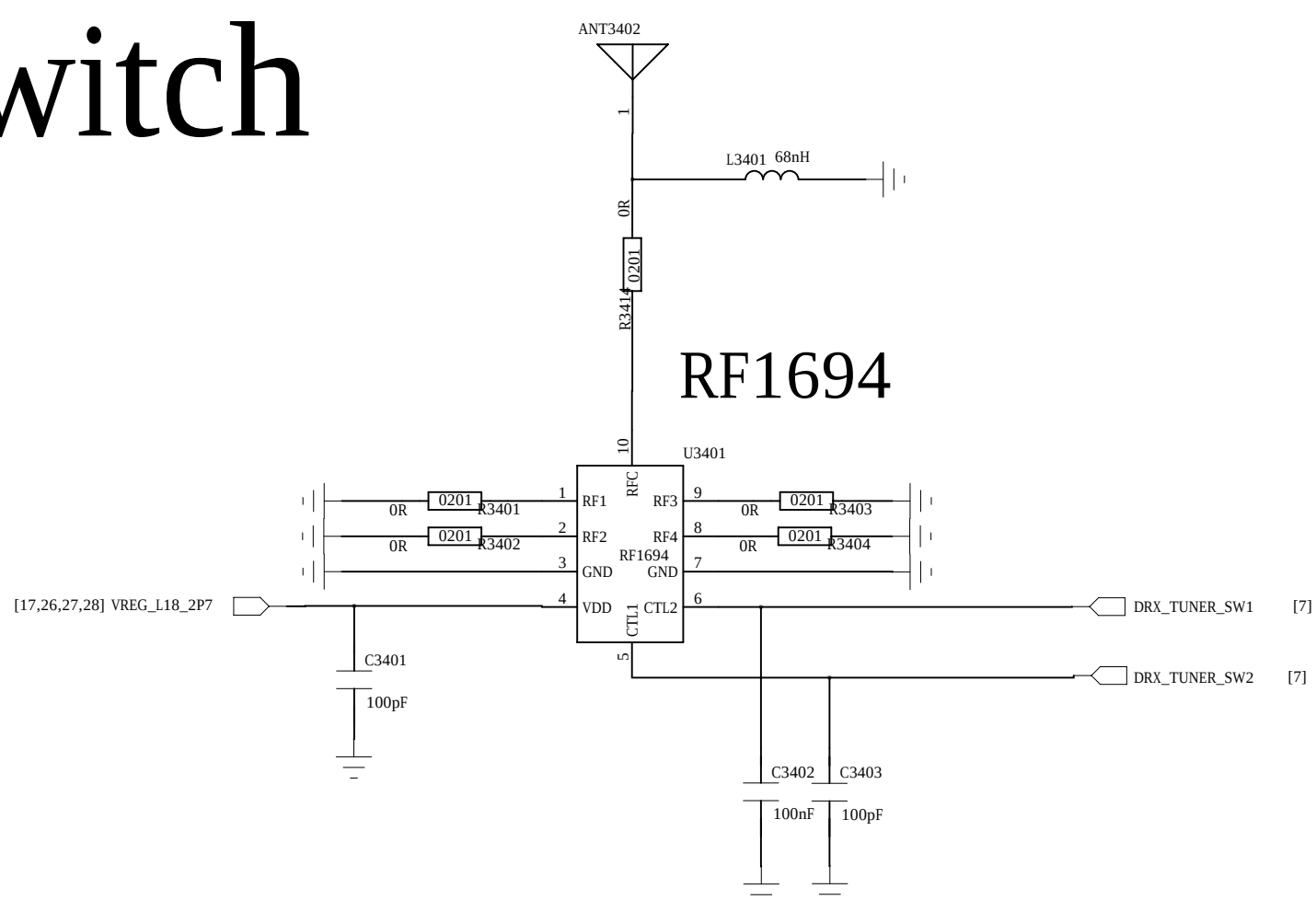
RFMD5428

Near to PA

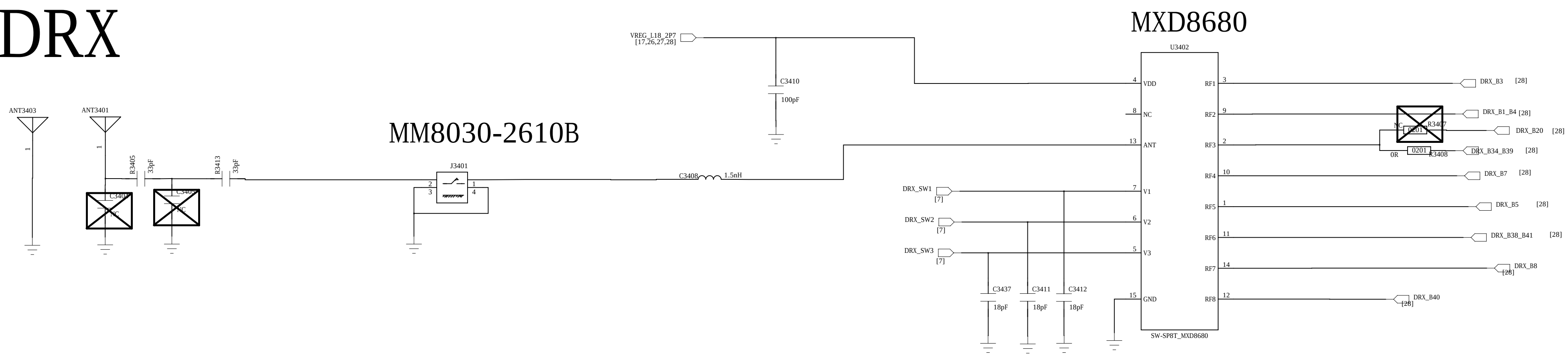


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TITLE: <Title>			
DRAWN: <Drawn By>	DATED: <Drawn Date>		
CHECKED: <Checked By>	DATED: <Checked Date>	CODE:	SIZE:
QUALITY CONTROL: <QC By>	DATED: <QC Date>	DRAWING NO.	
RELEASED: <Released By>	DATED: <Release Date>	REV:	
		<Code> A0 <Drawing Number>Revision>	
SCALE: <Scale>		SHEET: 20 31	

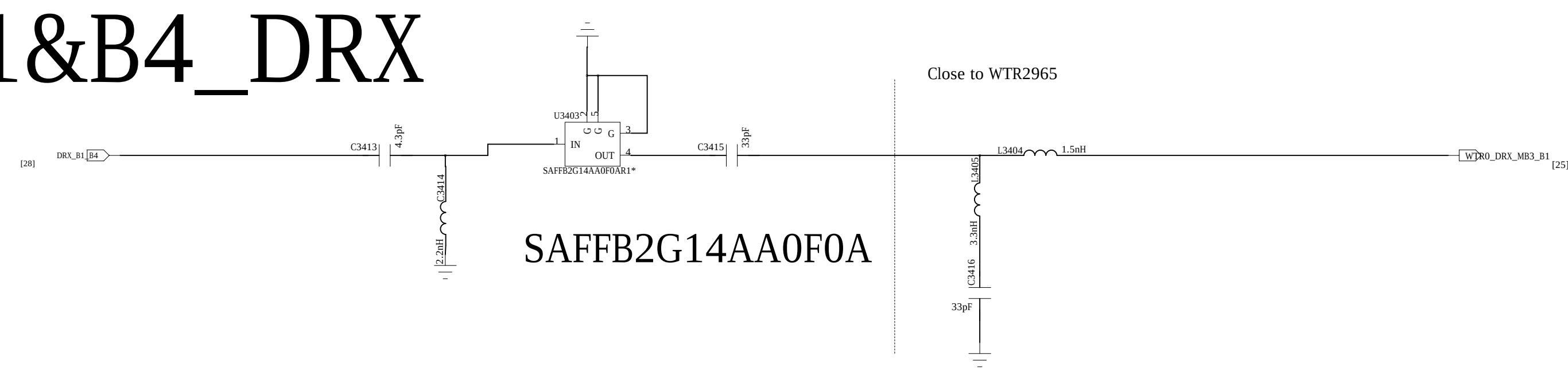
ANT Switch



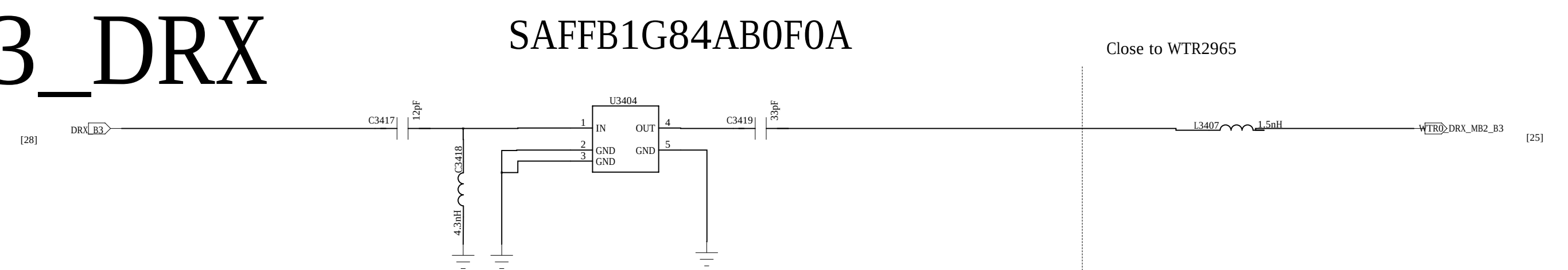
DRX



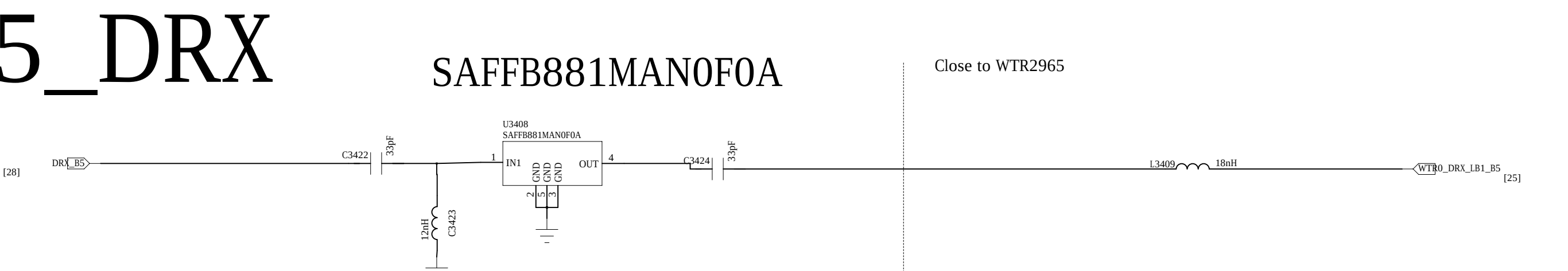
B1&B4_DRX



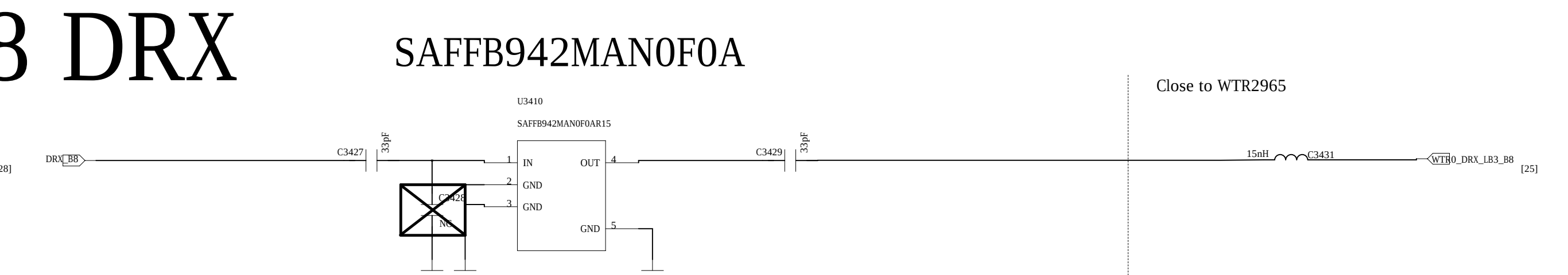
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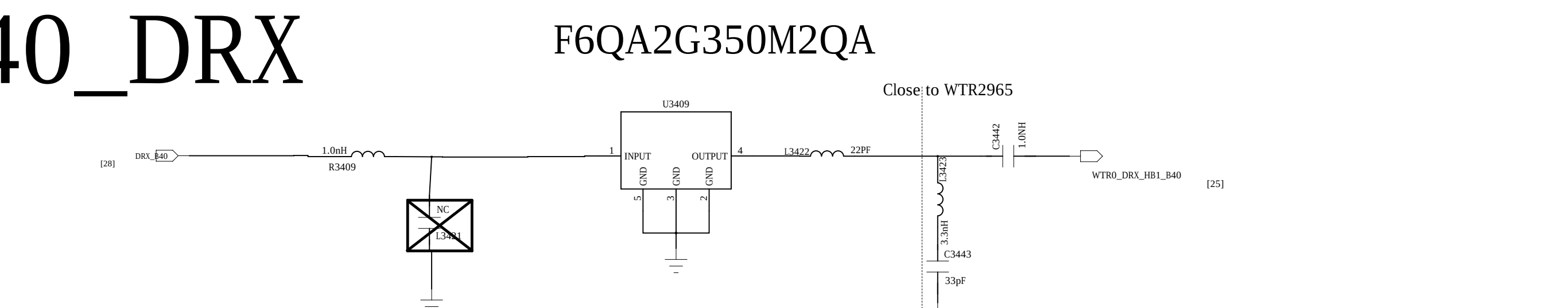
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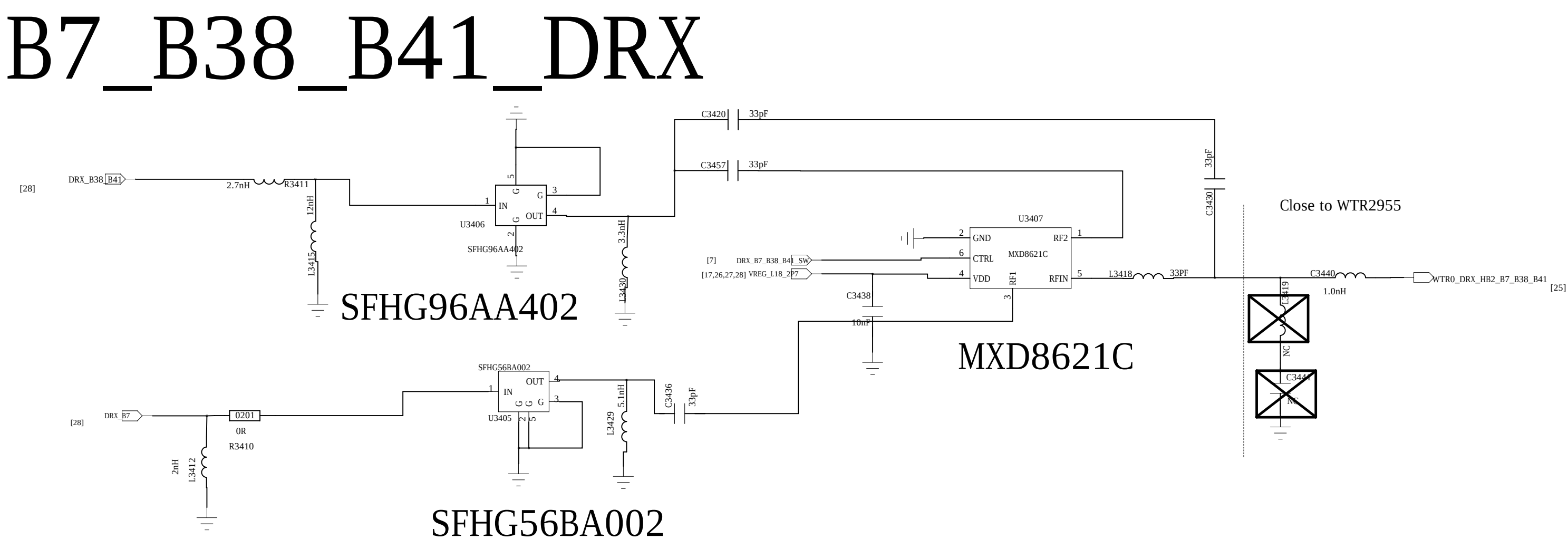
B8 DRX



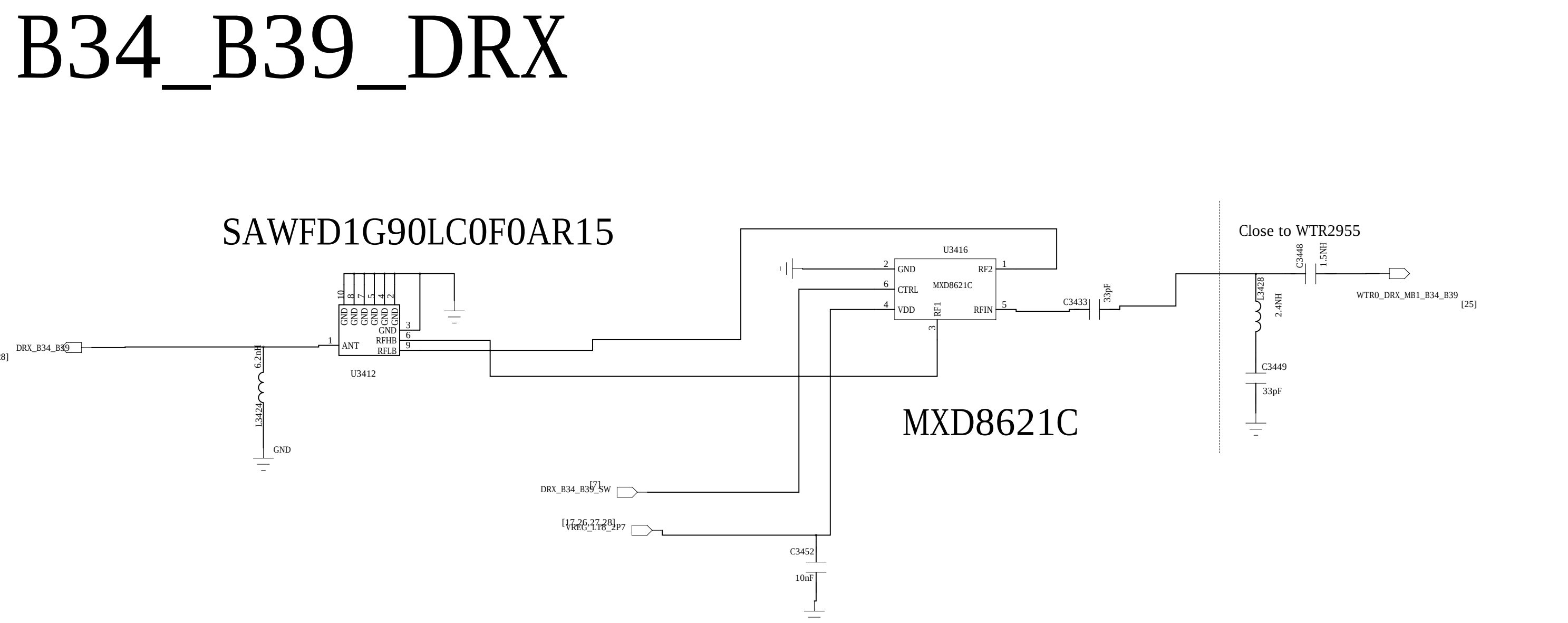
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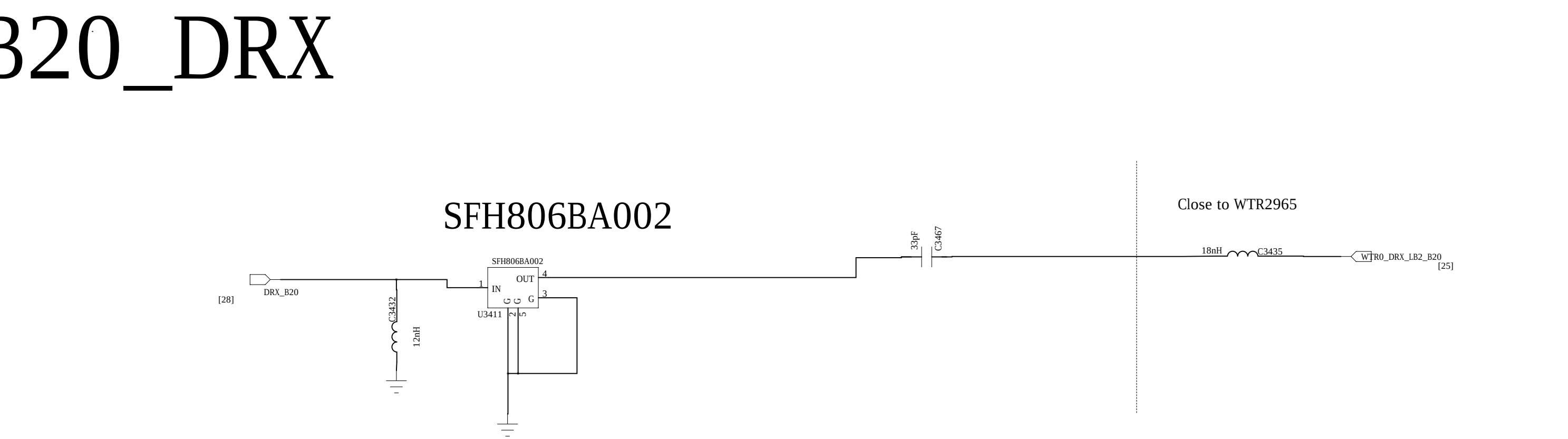
B7_B38_B41_DRX



B34_B39_DRX

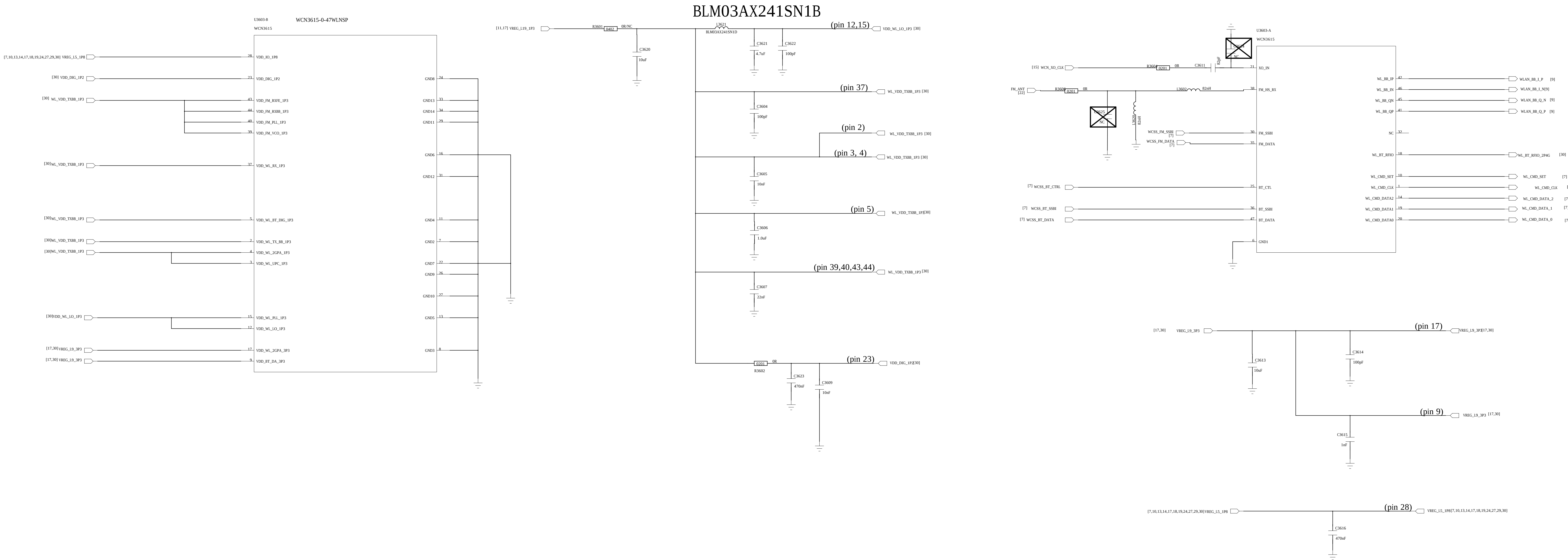


B20_DRX

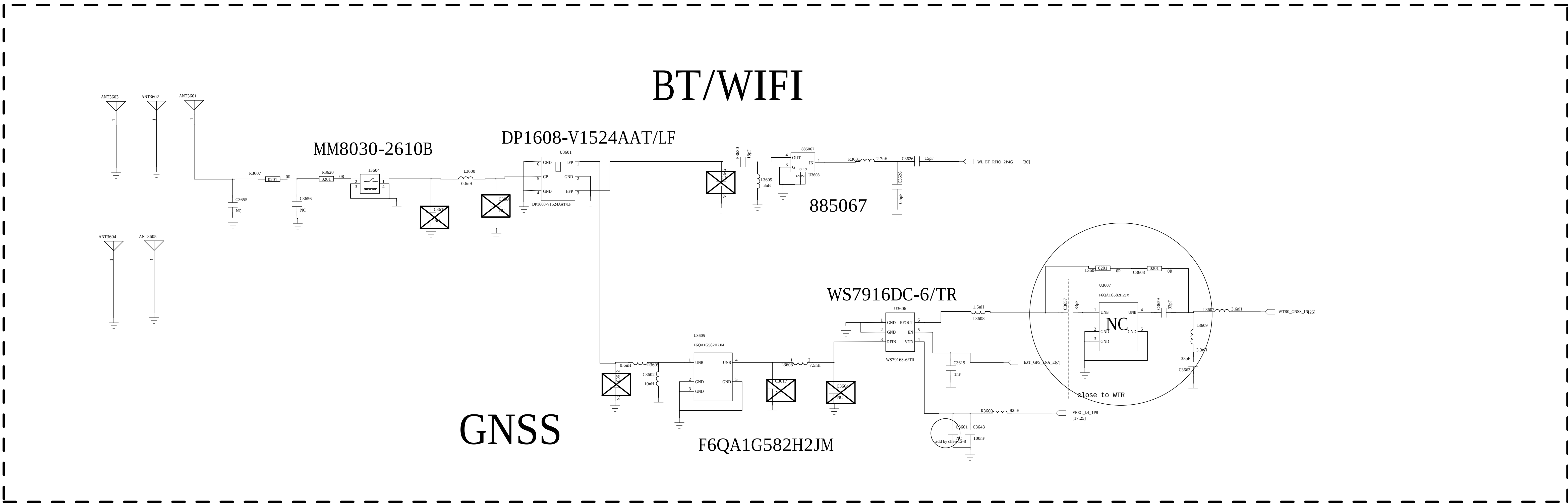


name of the schematic	37_WCN3680B
name of the board	GOHAN MAIN PCB
author	ZHAOLEI
number of page	34
total of pages	38
last modified date	2015.11.20
revision number	REV.A

WCN



BT/WIFI



GNSS

COMPANY:		<Company Name>	
TITLE:		24_WCN3680B	
CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A0	<Drawing Number>	<Revision>
DRAWN:	DATE:	CHECKED:	DATE:
LiuMeigra	<Drawn Date>	Liu	<Checked Date>
SCALE:		SHEET 30 of 31	

Change list

- 0519
- 1.SIM;”»»³ÉÍ¬D6
- 2.Ôø¼0¶||ú»ú²¿·Ôμ¿Â·

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Sheet		
Size	Name	Rev
Date: _____ Sheet _____ of _____		